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1 Basic

1.1 Default Code [81fabd]

```
#include <bits/stdc++.h>
#define For(i, a, b) for (int i = a; i <= b; i++)
#define Forr(i, a, b) for (int i = a; i >= b; i--)
#define F first
#define S second
#define fs F
#define sc S
#define iter(x) x.begin(), x.end()
#define sz(x) ((int)x.size())
using namespace std;
using ll = long long;
using pii = pair<int, int>;
using pll = pair<ll, ll>;

int32_t main() {
    ios::sync_with_stdio(false);
    cin.tie(0); // ^= -w -^=
}
```

1.2 Debugger [0d7909]

```
#ifdef zisk
template<class T> void _d(T &&x) {
    if constexpr (ranges::range<T> &&
        !is_convertible_v<T, string_view>) {
        cerr << "{ "; for (auto &&i:x) _d(i); cerr << " } ";
    } else if constexpr (requires { get<0>(x); }) {
        cerr << "( ";
        apply([&](auto &&...a){ (_d(a), ...); }, x);
        cerr << " ) ";
    } else cerr << x << " ";
} // ranges::subrange(L, r)
void _db() { cerr << "\e[0m\n"; }
void _db(auto &&...a) {
    cerr << "\e[1;33m"; (_d(a), ...); _db(); }
#define debug(...) _db(#__VA_ARGS__, ":", __VA_ARGS__)
#define safe_db(__PRETTY_FUNCTION__, __LINE__, "safe")
#else
#define safe void()
#define debug(...) void()
#endif
```

1.3 Fast IO [604966]

```
#pragma GCC optimize("Ofast,inline,unroll-loops")
#pragma GCC target("bmi,bmi2,lzcnt,popcnt,avx2,sse,sse2,sse3,ssse3,sse4,popcnt,abm,mmx,avx,tune=native")
#include<unistd.h>
char OB[65536]; int OP;
inline char RC() {
    static char buf[65536], *p = buf, *q = buf;
    return p == q && (q = (p = buf) + read(0, buf, 65536))
        == buf ? -1 : *p++;
}
inline int R() {
    static char c;
    while((c = RC()) < '0'); int a = c ^ '0';
    while((c = RC()) >= '0') a *= 10, a += c ^ '0';
    return a;
}
inline void W(int n) {
    static char buf[12], p;
    if (n == 0) OB[OP++] = '0'; p = 0;
    while (n) buf[p++] = '0' + (n % 10), n /= 10;
    for (--p; p >= 0; --p) OB[OP++] = buf[p];
    if (OP > 65520) write(1, OB, OP), OP = 0;
}
```

1.4 vimrc [e7a474]

```
sy on
se nu rnu so=8 bs=2 sw=4 ts=4 sts=4 hls ls=2 cin acd
mouse=a et
map <F9> :w<bar>!g++ "%" -o %:r -std=c++20 -Wall -
Wextra -Wshadow -Wconversion -O2 -Dzisk -g -
fsanitize=address,undefined<CR>
map <F8> :!./%:r<CR>
inoremap {<CR> {<CR>}<ESC>O
ca Hash w !cpp -dD -P -fpreprocessed \ tr -d '[:space
:]' \ md5sum \ cut -c-6
inoremap fj <ESC>
vnoremap fj <ESC>
// -D_GLIBCXX_ASSERTIONS, -D_GLIBCXX_DEBUG
```

2 Graph

2.1 SCC [79ed0d]

```
int BuildScc(int n, const auto &adj, const auto &radj,
    auto &scc) { // 0-indexed, returns # of SCCs
    vector<int> stk;
    int cnt = 0;
    auto dfs1 = [&](auto &self, int x) -> void {
        scc[x] = -1;
        for (auto i : adj[x])
            if (scc[i] == 0) self(self, i);
        stk.emplace_back(x);
    };
    auto dfs2 = [&](auto &self, int x) -> void {
        scc[x] = cnt;
        for (auto i : radj[x])
            if (scc[i] == -1) self(self, i);
    };
}
```

```

};
fill(scc, scc + n, 0);
For(i, 0, n - 1) if (scc[i] == 0) dfs1(dfs1, i);
for (auto i : stk | views::reverse)
    if (scc[i] == -1) dfs2(dfs2, i), cnt++;
return cnt;
}
// 2-SAT: alloc 2n vertices x0, x1
// x or y .. x0 -> y1, y0 -> x1
// forced x .. x0 -> x1
// no sol if scc[x0] == scc[x1]
// else, assign with the one with larger SCC id

```

2.2 BCC [f1ccc5]

```

struct BCC { // 0-based, allow multi edges but not allow
    loops
    int n, m, cnt = 0;
    // n:|V|, m:|E|, cnt:#bcc
    // bcc i : vertices bcc_v[i] and edges bcc_e[i]
    vector<vector<int>> bcc_v, bcc_e;
    vector<vector<pii>> g; // original graph
    vector<pii> edges; // 0-based
    BCC(int _n, vector<pii> _edges):
        n(_n), m(sz(_edges)), g(_n), edges(_edges){
        for(int i = 0; i < m; i++){
            auto [u, v] = edges[i];
            g[u].pb(pii(v, i)); g[v].pb(pii(u, i));
        }
    }
    void make_bcc(){ bcc_v.pb(); bcc_e.pb(); cnt++; }
    // modify these if you need more information
    void add_v(int v){ bcc_v.back().pb(v); }
    void add_e(int e){ bcc_e.back().pb(e); }
    void build(){
        vector<int> dfn(n, -1), low(n, -1), stk;
        vector<vector<int>> up(n);
        int ts = 0;
        auto _dfs = [&](auto dfs, int now, int par, int pe)
            -> void{
            if(pe != -1) up[now].pb(pe);
            in[now] = low[now] = ts++;
            stk.pb(now);
            for(auto [v, e] : g[now]){
                if(e == pe) continue;
                if(in[v] != -1){
                    if(in[v] < in[now]) up[now].pb(e);
                    low[now] = min(low[now], in[v]);
                    continue;
                }
                dfs(dfs, v, now, e);
                low[now] = min(low[now], low[v]);
            } // For Edge BCC, replace to low[now] == in[now]
                (for any now)
            if((now != par && low[now] >= in[par]) || (now ==
                par && sz(g[now]) == 0)){
                make_bcc();
                for(int v = stk.back(); v = stk.back()){
                    stk.pop_back(), add_v(v);
                    for(int e : up[v]) add_e(e);
                    if(v == now) break;
                }
                if(now != par) add_v(par);
            }
        };
        for(int i = 0; i < n; i++)
            if(in[i] == -1) _dfs(_dfs, i, i, -1);
    }
};

```

2.3 Centroid Decomposition [77ce50]

```

struct CenDecomp {
    int n, mxdep;
    vector<int> par, dep;
    CenDecomp(int _n, const auto &adj): n(_n), mxdep(0),
        par(n, -1), dep(n, -1) {
        vector<pii> szs(n); int ns = 0;
        auto getsz = [&](auto &self, int v, int p) -> int {
            int sz = 1;
            for (auto i : adj[v]) if (i != p && dep[i] == -1)
                sz += self(self, i, v);
        };
    }
};

```

```

        szs[ns++] = {sz, v};
        return sz;
    };
    auto dfs = [&](auto &self, int v, int p, int d) ->
        void {
            int tot = getsz(getsz, v, v), j = 0;
            while (szs[j].F <= tot / 2) j++;
            v = szs[j].S; ns = 0;
            par[v] = p; dep[v] = d;
            mxdep = max(mxdep, d + 1);
            for (auto i : adj[v]) if (dep[i] == -1)
                self(self, i, v, d + 1);
        };
    dfs(dfs, 0, -1, 0);
} // adj could be vector<vector> or vector[]
}; // 0-indexed nodes, par[root] = -1, dep[root] = 0

```

2.4 Count Cycles [c7e8f2]

```

// ord = sort by deg decreasing, rk[ord[i]] = i
// D[i] = edge point from rk small to rk big
for (int x : ord) { // c3
    for (int y : D[x]) vis[y] = 1;
    for (int y : D[x]) for (int z : D[y]) c3 += vis[z];
    for (int y : D[x]) vis[y] = 0;
}
for (int x : ord) { // c4
    for (int y : D[x]) for (int z : adj[y])
        if (rk[z] > rk[x]) c4 += vis[z]++;
    for (int y : D[x]) for (int z : adj[y])
        if (rk[z] > rk[x]) --vis[z];
} // both are O(M*sqrt(M)), test @ 2022 CCPC guangzhou

```

2.5 DirectedMST [d3eb5f]

```

using D = int;
struct edge {
    int u, v; D w;
};
// 0-based, return index of edges
vector<int> dmst(vector<edge> &e, int n, int root) {
    using T = pair<D, int>;
    using PQ = pair<priority_queue<T, vector<T>,
        greater<T>>, D>;
    auto push = [](PQ &pq, T v) {
        pq.first.emplace(v.first - pq.second, v.second);
    };
    auto top = [](const PQ &pq) -> T {
        auto r = pq.first.top();
        return {r.first + pq.second, r.second};
    };
    auto join = [&push, &top](PQ &a, PQ &b) {
        if (a.first.size() < b.first.size()) swap(a, b);
        while (!b.first.empty())
            push(a, top(b)), b.first.pop();
    };
    vector<PQ> h(n * 2);
    for (int i = 0; i < e.size(); ++i)
        push(h[e[i].v], {e[i].w, i});
    vector<int> a(n * 2), v(n * 2, -1), pa(n * 2, -1), r(
        n * 2);
    iota(a.begin(), a.end(), 0);
    auto o = [&](int x) { int y;
        for (y = x; a[y] != y; y = a[y]);
        for (int ox = x; x != y; ox = x)
            x = a[x], a[ox] = y;
        return y;
    };
    v[root] = n + 1;
    int pc = n;
    for (int i = 0; i < n; ++i) if (v[i] == -1) {
        for (int p = i; v[p] == -1 || v[p] == i; p = o(e[r[
            p]].u)) {
            if (v[p] == i) {
                int q = p; p = pc++;
                do {
                    h[q].second = -h[q].first.top().first;
                    join(h[pa[q]] = a[q] = p], h[q]);
                } while ((q = o(e[r[q]].u)) != p);
            }
            v[p] = i;
            while (!h[p].first.empty() && o(e[top(h[p]).
                second].u) == p)
        }
    }
}

```

```

        h[p].first.pop();
        r[p] = top(h[p]).second;
    }
}
vector<int> ans;
for (int i = pc - 1; i >= 0; i--)
    if (i != root && v[i] != n) {
        for (int f = e[r[i]].v; f != -1 && v[f] != n; f =
            pa[f]) v[f] = n;
        ans.pb(r[i]);
    }
return ans;
}

```

2.6 Dominator Tree [6378d5]

```

struct Dominator_tree {
    int n, id;
    vector<vector<int>> adj, radj, bucket;
    vector<int> sdom, dom, vis, rev, par, rt, mn;
    Dominator_tree (int _n) : n(_n), id(0) {
        adj.resize(n), radj.resize(n), bucket.resize(n);
        sdom.resize(n), dom.resize(n, -1), vis.resize(n,
            -1);
        rev.resize(n), rt.resize(n), mn.resize(n), par.
            resize(n);
    }
    void add_edge(int u, int v) {adj[u].pb(v);}
    int query(int v, bool x) {
        if (rt[v] == v) return x ? -1 : v;
        int p = query(rt[v], true);
        if (p == -1) return x ? rt[v] : mn[v];
        if (sdom[mn[v]] > sdom[mn[rt[v]]]) mn[v] = mn[rt[v]
            ];
        rt[v] = p;
        return x ? p : mn[v];
    }
    void dfs(int v) {
        vis[v] = id, rev[id] = v;
        rt[id] = mn[id] = sdom[id] = id, id++;
        for (int u : adj[v]) {
            if (vis[u] == -1) dfs(u), par[vis[u]] = vis[v];
            radj[vis[u]].pb(vis[v]);
        }
    }
    void build(int s) {
        dfs(s);
        for (int i = id - 1; ~i; --i) {
            for (int u : radj[i]) {
                sdom[i] = min(sdom[i], sdom[query(u, false)]);
            }
            if (i) bucket[sdom[i]].pb(i);
            for (int u : bucket[i]) {
                int p = query(u, false);
                dom[u] = sdom[p] == i ? i : p;
            }
            if (i) rt[i] = par[i];
        }
        vector<int> res(n, -1);
        for (int i = 1; i < id; ++i) {
            if (dom[i] != sdom[i]) dom[i] = dom[dom[i]];
        }
        for (int i = 1; i < id; ++i) res[rev[i]] = rev[dom[
            i]];
        res[s] = s;
        dom = res;
    } // dom[]: parent on dom. tree, -1 if not reachable
};

```

2.7 Heavy Light Decomposition [372958]

```

vector<int> adj[N];
struct HLD {
    // 0-index
    vector<int> dep, pt, hd, idx, sz, par, vis;
    int n, _t;
    HLD () = default;
    HLD (int _n) : n(_n) {
        pt.assign(n, -1), hd.assign(n, -1), par.assign(n, -1);
        idx.assign(n, 0), sz.assign(n, 0), dep.assign(n, 0),
            vis.assign(n, 0);
        _t = 0;
    }
}

```

```

for (int i = 0; i < n; ++i) if (!vis[i]) {
    dfs1(i, -1);
    dfs2(i, -1, 0);
}
}
void dfs1(int v, int pa) {
    par[v] = pa;
    dep[v] = ~pa ? dep[pa] + 1 : 0;
    sz[v] = vis[v] = 1;
    for (int u : adj[v]) if (u != pa) {
        dfs1(u, v);
        if (pt[v] == -1 || sz[pt[v]] < sz[u])
            pt[v] = u;
        sz[v] += sz[u];
    }
}
void dfs2(int v, int pa, int h) {
    if (v == -1)
        return;
    idx[v] = _t++, hd[v] = h;
    dfs2(pt[v], v, h);
    for (int u : adj[v]) if (u != pa && u != pt[v]) {
        dfs2(u, v, u);
    }
}
void modify(int u, int v) {
    while (hd[u] != hd[v]) {
        if (dep[hd[u]] < dep[hd[v]])
            swap(u, v);
        // range [idx[hd[u]], idx[u] + 1)
        u = par[hd[u]];
    }
    if (dep[u] < dep[v])
        swap(u, v);
    // range [idx[v], idx[u] + 1)
}
int query(int u, int v) {
    int ans = 0;
    while (hd[u] != hd[v]) {
        if (dep[hd[u]] < dep[hd[v]])
            swap(u, v);
        // range [idx[hd[u]], idx[u] + 1)
        u = par[hd[u]];
    }
    if (dep[u] < dep[v])
        swap(u, v);
    // range [idx[v], idx[u] + 1)
    return ans;
}
};

```

2.8 Virtual Tree [dcbe4f]

```

// need lca
vector<int> _g[N], stk;
int st[N], ed[N];
void solve(vector<int> v) {
    auto cmp = [&](int x, int y) {return st[x] < st[y];};
    sort(all(v), cmp);
    int sz = v.size();
    for (int i = 0; i < sz - 1; ++i)
        v.pb(lca(v[i], v[i + 1]));
    sort(all(v), cmp);
    v.resize(unique(all(v)) - v.begin());
    stk.clear(), stk.pb(v[0]);
    for (int i = 1; i < v.size(); ++i) {
        int x = v[i];
        while (ed[stk.back()] < ed[x]) stk.pop_back();
        _g[stk.back()].pb(x), stk.pb(x);
    }
    // do something
    for (int i : v) _g[i].clear();
}

```

2.9 Vizing [fa4b32]

```

namespace vizing { // returns edge coloring in adjacent
    matrix G, 1 - based
    const int N = 105;
    int C[N][N], G[N][N], X[N], vst[N], n;
    void init(int _n) { n = _n;
        for (int i = 0; i <= n; ++i)
    }
}

```

```

    for (int j = 0; j <= n; ++j)
        C[i][j] = G[i][j] = 0;
}
void solve(vector<pii> &E) {
    auto update = [&](int u)
    { for (X[u] = 1; C[u][X[u]]; ++X[u]); };
    auto color = [&](int u, int v, int c) {
        int p = G[u][v];
        G[u][v] = G[v][u] = c;
        C[u][c] = v, C[v][c] = u;
        C[u][p] = C[v][p] = 0;
        if (p) X[u] = X[v] = p;
        else update(u), update(v);
        return p;
    };
    auto flip = [&](int u, int c1, int c2) {
        int p = C[u][c1];
        swap(C[u][c1], C[u][c2]);
        if (p) G[u][p] = G[p][u] = c2;
        if (!C[u][c1]) X[u] = c1;
        if (!C[u][c2]) X[u] = c2;
        return p;
    };
    fill_n(X + 1, n, 1);
    for (int t = 0; t < E.size(); ++t) {
        int u = E[t].F, v0 = E[t].S, v = v0, c0 = X[u], c =
            c0, d;
        vector<pii> L;
        fill_n(vst + 1, n, 0);
        while (!G[u][v0]) {
            L.emplace_back(v, d = X[v]);
            if (!C[v][c]) for (int a = (int)L.size() - 1; a
                >= 0; --a) c = color(u, L[a].F, c);
            else if (!C[u][d]) for (int a = (int)L.size() -
                1; a >= 0; --a) color(u, L[a].F, L[a].S);
            else if (vst[d]) break;
            else vst[d] = 1, v = C[u][d];
        }
        if (!G[u][v0]) {
            for (; v; v = flip(v, c, d), swap(c, d));
            if (int a; C[u][c0]) {
                for (a = (int)L.size() - 2; a >= 0 && L[a].S !=
                    c; --a);
                for (; a >= 0; --a) color(u, L[a].F, L[a].S);
            }
            else --t;
        }
    }
}
// namespace vizing

```

2.10 Maximum Clique Dynamic [6dde09]

```

const int N = 150;
struct MaxClique { // Maximum Clique
    bitset<N> a[N], cs[N];
    int ans, sol[N], q, cur[N], d[N], n;
    void init(int _n) {
        n = _n;
        for (int i = 0; i < n; i++) a[i].reset();
    }
    void addEdge(int u, int v) { a[u][v] = a[v][u] = 1; }
    void csort(vector<int> &r, vector<int> &c) {
        int mx = 1, km = max(ans - q + 1, 1), t = 0,
            m = r.size();
        cs[1].reset(), cs[2].reset();
        for (int i = 0; i < m; i++) {
            int p = r[i], k = 1;
            while ((cs[k] & a[p]).count()) k++;
            if (k > mx) mx++, cs[mx + 1].reset();
            cs[k][p] = 1;
            if (k < km) r[t++] = p;
        }
        c.resize(m);
        if (t) c[t - 1] = 0;
        for (int k = km; k <= mx; k++)
            for (int p = cs[k]._Find_first(); p < N;
                p = cs[k]._Find_next(p))
                r[t] = p, c[t] = k, t++;
    }
    void dfs(vector<int> &r, vector<int> &c, int l,
        bitset<N> mask) {

```

```

        while (!r.empty()) {
            int p = r.back();
            r.pop_back(), mask[p] = 0;
            if (q + c.back() <= ans) return;
            cur[q++] = p;
            vector<int> nr, nc;
            bitset<N> nmask = mask & a[p];
            for (int i : r)
                if (a[p][i]) nr.push_back(i);
            if (!nr.empty()) {
                if (1 < 4) {
                    for (int i : nr)
                        d[i] = (a[i] & nmask).count();
                    sort(nr.begin(), nr.end(),
                        [&](int x, int y) { return d[x] > d[y]; });
                }
                csort(nr, nc), dfs(nr, nc, l + 1, nmask);
            } else if (q > ans) ans = q, copy_n(cur, q, sol);
            c.pop_back(), q--;
        }
    }
    int solve(bitset<N> mask = bitset<N>()
        string(N, '1')) { // vertex mask
        vector<int> r, c;
        ans = q = 0;
        for (int i = 0; i < n; i++)
            if (mask[i]) r.push_back(i);
        for (int i = 0; i < n; i++)
            d[i] = (a[i] & mask).count();
        sort(r.begin(), r.end(),
            [&](int i, int j) { return d[i] > d[j]; });
        csort(r, c), dfs(r, c, 1, mask);
        return ans; // sol[0 ~ ans-1]
    }
} graph;

```

2.11 Minimum Steiner Tree [c87e6f]

```

// O(V 3^T + V^2 2^T)
struct SteinerTree { // 0-base
    static const int T = 10, N = 105, INF = 1e9;
    int n, dst[N][N], dp[1 << T][N], tdst[N];
    int vcost[N]; // the cost of vertices
    void init(int _n) {
        n = _n;
        for (int i = 0; i < n; ++i) {
            for (int j = 0; j < n; ++j) dst[i][j] = INF;
            dst[i][i] = vcost[i] = 0;
        }
    }
    void add_edge(int ui, int vi, int wi) {
        dst[ui][vi] = min(dst[ui][vi], wi);
    }
    void shortest_path() {
        for (int k = 0; k < n; ++k)
            for (int i = 0; i < n; ++i)
                for (int j = 0; j < n; ++j)
                    dst[i][j] =
                        min(dst[i][j], dst[i][k] + dst[k][j]);
    }
    int solve(const vector<int> &ter) {
        shortest_path();
        int t = sz(ter);
        for (int i = 0; i < (1 << t); ++i)
            for (int j = 0; j < n; ++j) dp[i][j] = INF;
        for (int i = 0; i < n; ++i) dp[0][i] = vcost[i];
        for (int msk = 1; msk < (1 << t); ++msk) {
            if (!(msk & (msk - 1))) {
                int who = __lg(msk);
                for (int i = 0; i < n; ++i)
                    dp[msk][i] =
                        vcost[ter[who]] + dst[ter[who]][i];
            }
            for (int i = 0; i < n; ++i)
                for (int submsk = (msk - 1) & msk; submsk =
                    (submsk - 1) & msk)
                    dp[msk][i] = min(dp[msk][i],
                        dp[submsk][i] + dp[msk ^ submsk][i] -
                            vcost[i]);
            for (int i = 0; i < n; ++i) {
                tdst[i] = INF;
                for (int j = 0; j < n; ++j)

```

```

        tdst[i] =
            min(tdst[i], dp[msk][j] + dst[j][i]);
    }
    for (int i = 0; i < n; ++i) dp[msk][i] = tdst[i];
}
int ans = INF;
for (int i = 0; i < n; ++i)
    ans = min(ans, dp[(1 << t) - 1][i]);
return ans;
}
};

```

2.12 Theory

$| \text{Maximum independent edge set} | = |V| - | \text{Minimum edge cover} |$
 $| \text{Maximum independent set} | = |V| - | \text{Minimum vertex cover} |$

3 Data Structure

3.1 Dynamic Line Hull [8ec1c7]

```

struct Line {
    mutable ll k, m, p;
    bool operator<(const Line& o) const { return k < o.k; }
    bool operator<(ll x) const { return p < x; }
};

struct LineContainer : multiset<Line, less<>> {
    static const ll inf = LLONG_MAX;
    ll div(ll a, ll b) {
        return a / b - ((a ^ b) < 0 && a % b);
    }
    bool isect(iterator x, iterator y) {
        if (y == end()) return x->p = inf, 0;
        if (x->k == y->k) x->p = x->m > y->m ? inf : -inf;
        else x->p = div(y->m - x->m, x->k - y->k);
        return x->p >= y->p;
    }
    void add(ll k, ll m) {
        auto z = insert({k, m, 0}), y = z++, x = y;
        while (isect(y, z)) z = erase(z);
        if (x != begin() && isect(--x, y)) isect(x, y = erase(y));
        while ((y = x) != begin() && (--x)->p >= y->p)
            isect(x, erase(y));
    }
    ll query(ll x) {
        assert(!empty());
        auto l = *lower_bound(x);
        return l.k * x + l.m;
    }
};

```

3.2 Leftist Tree [473c12]

```

struct node {
    ll rk, data, sz, sum;
    node *l, *r;
    node(ll k) : rk(0), data(k), sz(1), l(0), r(0), sum(k) {}
};

ll sz(node *p) { return p ? p->sz : 0; }
ll rk(node *p) { return p ? p->rk : -1; }
ll sum(node *p) { return p ? p->sum : 0; }

node *merge(node *a, node *b) {
    if (!a || !b) return a ? a : b;
    if (a->data < b->data) swap(a, b);
    a->r = merge(a->r, b);
    if (rk(a->r) > rk(a->l)) swap(a->r, a->l);
    a->rk = rk(a->r) + 1, a->sz = sz(a->l) + sz(a->r) + 1;
    a->sum = sum(a->l) + sum(a->r) + a->data;
    return a;
}

void pop(node *&o) {
    node *tmp = o;
    o = merge(o->l, o->r);
    delete tmp;
}

```

3.3 Link Cut Tree [87ade4]

```

// weighted subtree size, weighted path max
struct LCT {
    int ch[N][2], pa[N], v[N], sz[N], sz2[N], w[N], mx[N], _id;
    // sz := sum of v in splay, sz2 := sum of v in virtual subtree
    // mx := max w in splay
    bool rev[N];
    LCT() : _id(1) {}
    int newnode(int _v, int _w) {
        int x = _id++;
        ch[x][0] = ch[x][1] = pa[x] = 0;
        v[x] = sz[x] = _v;
        sz2[x] = 0;
        w[x] = mx[x] = _w;
        rev[x] = false;
        return x;
    }
    void pull(int i) {
        sz[i] = v[i] + sz2[i];
        mx[i] = w[i];
        if (ch[i][0])
            sz[i] += sz[ch[i][0]], mx[i] = max(mx[i], mx[ch[i][0]]);
        if (ch[i][1])
            sz[i] += sz[ch[i][1]], mx[i] = max(mx[i], mx[ch[i][1]]);
    }
    void push(int i) {
        if (rev[i]) reverse(ch[i][0]), reverse(ch[i][1]), rev[i] = false;
    }
    void reverse(int i) {
        if (!i) return;
        swap(ch[i][0], ch[i][1]);
        rev[i] ^= true;
    }
    bool isrt(int i) { // rt of splay
        if (!pa[i]) return true;
        return ch[pa[i]][0] != i && ch[pa[i]][1] != i;
    }
    void rotate(int i) {
        int p = pa[i], x = ch[p][1] == i, c = ch[i][!x], gp = pa[p];
        if (ch[gp][0] == p) ch[gp][0] = i;
        else if (ch[gp][1] == p) ch[gp][1] = i;
        pa[i] = gp, ch[i][!x] = p, pa[p] = i;
        ch[p][x] = c, pa[c] = p;
        pull(p), pull(i);
    }
    void splay(int i) {
        vector<int> anc;
        anc.push_back(i);
        while (!isrt(anc.back())) anc.push_back(pa[anc.back()]);
        while (!anc.empty()) push(anc.back()), anc.pop_back();
        while (!isrt(i)) {
            int p = pa[i];
            if (!isrt(p)) rotate(ch[p][1] == i ^ ch[pa[p]][1] == p ? i : p);
            rotate(i);
        }
    }
    void access(int i) {
        int last = 0;
        while (i) {
            splay(i);
            if (ch[i][1])
                sz2[i] += sz[ch[i][1]];
            sz2[i] -= sz[last];
            ch[i][1] = last;
            pull(i), last = i, i = pa[i];
        }
    }
    void makert(int i) {
        access(i), splay(i), reverse(i);
    }
    void link(int i, int j) {
        // assert(findrt(i) != findrt(j));
    }
}

```

```

    makert(i);
    makert(j);
    pa[i] = j;
    sz2[j] += sz[i];
    pull(j);
}
void cut(int i, int j) {
    makert(i), access(j), splay(i);
    // assert(sz[i] == 2 && ch[i][1] == j);
    ch[i][1] = pa[j] = 0, pull(i);
}
int findrt(int i) {
    access(i), splay(i);
    while (ch[i][0]) push(i), i = ch[i][0];
    splay(i);
    return i;
}
};

```

3.4 Splay Tree [e9029a]

```

struct Splay {
    int pa[N], ch[N][2], sz[N], rt, _id;
    ll v[N];
    Splay() {}
    void init() {
        rt = 0, pa[0] = ch[0][0] = ch[0][1] = -1;
        sz[0] = 1, v[0] = inf;
    }
    int newNode(int p, int x) {
        int id = _id++;
        v[id] = x, pa[id] = p;
        ch[id][0] = ch[id][1] = -1, sz[id] = 1;
        return id;
    }
    void rotate(int i) {
        int p = pa[i], x = ch[p][1] == i, gp = pa[p], c =
            ch[i][!x];
        sz[p] -= sz[i], sz[i] += sz[p];
        if (~c) sz[p] += sz[c], pa[c] = p;
        ch[p][x] = c, pa[p] = i;
        pa[i] = gp, ch[i][!x] = p;
        if (~gp) ch[gp][ch[gp][1] == p] = i;
    }
    void splay(int i) {
        while (~pa[i]) {
            int p = pa[i];
            if (~pa[p]) rotate(ch[pa[p]][1] == p ^ ch[p][1]
                == i ? i : p);
            rotate(i);
        }
        rt = i;
    }
    int lower_bound(int x) {
        int i = rt, last = -1;
        while (true) {
            if (v[i] == x) return splay(i), i;
            if (v[i] > x) {
                last = i;
                if (ch[i][0] == -1) break;
                i = ch[i][0];
            }
            else {
                if (ch[i][1] == -1) break;
                i = ch[i][1];
            }
        }
        splay(i);
        return last; // -1 if not found
    }
    void insert(int x) {
        int i = lower_bound(x);
        if (i == -1) {
            // assert(ch[rt][1] == -1);
            int id = newNode(rt, x);
            ch[rt][1] = id, ++sz[rt];
            splay(id);
        }
        else if (v[i] != x) {
            splay(i);
            int id = newNode(rt, x), c = ch[rt][0];
            ch[rt][0] = id;
        }
    }
};

```

```

        ch[id][0] = c;
        if (~c) pa[c] = id, sz[id] += sz[c];
        ++sz[rt];
        splay(id);
    }
};

```

4 Flow/Matching

4.1 Hopcroft Karp [4b930f]

```

struct HopcroftKarp {
    const int INF = 1 << 30;
    vector<int> adj[N];
    int match[N], dis[N], v, n, m;
    bool matched[N], vis[N];
    bool dfs(int x) { // dffdc
        vis[x] = true;
        for (int y : adj[x])
            if (match[y] == -1 || (dis[match[y]] == dis[x] +
                1 && !vis[match[y]] && dfs(match[y]))) {
                match[y] = x, matched[x] = true;
                return true;
            }
        return false;
    }
    bool bfs() { // c44dc
        memset(dis, -1, sizeof(int) * n);
        queue<int> q;
        for (int x = 0; x < n; ++x) if (!matched[x])
            dis[x] = 0, q.push(x);
        int mx = INF;
        while (!q.empty()) {
            int x = q.front(); q.pop();
            for (int y : adj[x]) {
                if (match[y] == -1) {
                    mx = dis[x];
                    break;
                }
                else if (dis[match[y]] == -1)
                    dis[match[y]] = dis[x] + 1, q.push(match[y]);
            }
        }
        return mx < INF;
    }
    int solve() { // 39e2af
        int res = 0;
        memset(match, -1, sizeof(int) * m);
        memset(matched, 0, sizeof(bool) * n);
        while (bfs()) {
            memset(vis, 0, sizeof(bool) * n);
            for (int x = 0; x < n; ++x) if (!matched[x])
                res += dfs(x);
        }
        return res;
    }
    void init(int _n, int _m) {
        n = _n, m = _m;
        for (int i = 0; i < n; ++i) adj[i].clear();
    }
    void add_edge(int x, int y) {
        adj[x].pb(y);
    }
};

```

4.2 Dinic [aa8522]

```

template <typename T> // 0-based,  $O(V^2E)$ , unit flow:  $O(\min(V^{2/3}E, E^{3/2}))$ , bipartite matching:  $O(\sqrt{VE})$ 
struct Dinic { // 0-based
    const T INF = numeric_limits<T>::max() / 2;
    struct edge { int to, rev; T cap, flow; };
    int n, s, t;
    vector<vector<edge>> g;
    vector<int> dis, cur;
    T dfs(int u, T cap) { // 09625c
        if (u == t || !cap) return cap;
        for (int &i = cur[u]; i < (int)g[u].size(); ++i) {
            edge &e = g[u][i];
            if (dis[e.to] == dis[u] + 1 && e.flow != e.cap) {

```

```

    T df = dfs(e.to, min(e.cap - e.flow, cap));
    if (df) {
        e.flow += df;
        g[e.to][e.rev].flow -= df;
        return df;
    } } }
dis[u] = -1;
return 0;
}
bool bfs() { // b06b3d
    fill(all(dis), -1);
    queue<int> q;
    q.push(s); dis[s] = 0;
    while (!q.empty()) {
        int v = q.front(); q.pop();
        for (auto &u : g[v])
            if (dis[u.to] == -1 && u.flow != u.cap) {
                q.push(u.to);
                dis[u.to] = dis[v] + 1;
            }
    }
    return dis[t] != -1;
}
T solve(int _s, int _t) { // 864a07
    s = _s; t = _t;
    T flow = 0, df;
    while (bfs()) {
        fill(all(cur), 0);
        while ((df = dfs(s, INF))) flow += df;
    }
    return flow;
}
void reset() {
    for (int i = 0; i < n; ++i)
        for (auto &j : g[i]) j.flow = 0;
}
void add_edge(int u, int v, T cap) {
    g[u].pb(edge{v, (int)g[v].size(), cap, 0});
    g[v].pb(edge{u, (int)g[u].size() - 1, 0, 0});
    //change g[v] to cap for undirected graphs
}
Dinic (int _n) : n(_n), g(n), dis(n), cur(n) {}
};

```

4.3 Min Cost Max Flow [e18ab8]

```

struct MCMF {
    const int INF = 1 << 30;
    struct edge {
        int v, f, c;
        edge(int _v, int _f, int _c) : v(_v), f(_f), c(_c) {}
    };
    vector<edge> E;
    vector<vector<int>> adj;
    vector<int> dis, pot, rt;
    int n, s, t;
    MCMF(int _n, int _s, int _t) : n(_n), s(_s), t(_t) {
        adj.resize(n);
    }
    void add_edge(int u, int v, int f, int c) {
        adj[u].pb(E.size(), E.pb(edge(v, f, c)));
        adj[v].pb(E.size(), E.pb(edge(u, 0, -c)));
    }
    bool SPFA() { // e9eac4
        rt.assign(n, -1), dis.assign(n, INF);
        vector<bool> vis(n, false);
        queue<int> q;
        q.push(s), dis[s] = 0, vis[s] = true;
        while (!q.empty()) {
            int v = q.front(); q.pop();
            vis[v] = false;
            for (int id : adj[v]) if (E[id].f > 0 && dis[E[id].v] > dis[v] + E[id].c + pot[v] - pot[E[id].v]) {
                dis[E[id].v] = dis[v] + E[id].c + pot[v] - pot[E[id].v];
                E[id].v, rt[E[id].v] = id;
                if (!vis[E[id].v]) vis[E[id].v] = true, q.push(E[id].v);
            }
        }
        return dis[t] != INF;
    }
};

```

```

}
bool dijkstra() { // ce8d3f
    rt.assign(n, -1), dis.assign(n, INF);
    priority_queue<pair<int, int>, vector<pair<int, int>>, greater<pair<int, int>>> pq;
    dis[s] = 0, pq.emplace(dis[s], s);
    while (!pq.empty()) {
        int d, v; tie(d, v) = pq.top(); pq.pop();
        if (dis[v] < d) continue;
        for (int id : adj[v]) if (E[id].f > 0 && dis[E[id].v] > dis[v] + E[id].c + pot[v] - pot[E[id].v]) {
            dis[E[id].v] = dis[v] + E[id].c + pot[v] - pot[E[id].v];
            E[id].v, rt[E[id].v] = id;
            pq.emplace(dis[E[id].v], E[id].v);
        }
    }
    return dis[t] != INF;
}
pair<int, int> runFlow() { // 9d16df
    pot.assign(n, 0);
    int cost = 0, flow = 0;
    bool fr = true;
    while ((fr ? SPFA() : dijkstra())) {
        for (int i = 0; i < n; ++i) {
            dis[i] += pot[i] - pot[s];
        }
        int add = INF;
        for (int i = t; i != s; i = E[rt[i] ^ 1].v) {
            add = min(add, E[rt[i]].f);
        }
        for (int i = t; i != s; i = E[rt[i] ^ 1].v) {
            E[rt[i]].f -= add, E[rt[i] ^ 1].f += add;
        }
        flow += add, cost += add * dis[t];
        fr = false;
        swap(dis, pot);
    }
    return make_pair(flow, cost);
}
};

```

4.4 Min Cost Circulation [ea0477]

```

template<typename F, typename C>
struct MinCostCirculation {
    struct ep { int to; F flow; C cost; };
    int n; vector<int> vis; int visc;
    vector<int> fa, fae; vector<vector<int>> g;
    vector<ep> e; vector<C> pi;
    MinCostCirculation(int n_) : n(n_), vis(n), visc(0),
        g(n), pi(n) {}
    void add_edge(int u, int v, F fl, C cs) {
        g[u].emplace_back((int)e.size());
        e.emplace_back(v, fl, cs);
        g[v].emplace_back((int)e.size());
        e.emplace_back(u, 0, -cs);
    }
    C phi(int x) {
        if (fa[x] == -1) return 0;
        if (vis[x] == visc) return pi[x];
        vis[x] = visc;
        return pi[x] = phi(fa[x]) - e[fae[x]].cost;
    }
    int lca(int u, int v) {
        for (; u != -1 || v != -1; swap(u, v)) if (u != -1) {
            if (vis[u] == visc) return u;
            vis[u] = visc; u = fa[u];
        }
        return -1;
    }
    void pushflow(int x, C &cost) {
        int v = e[x ^ 1].to, u = e[x].to; ++visc;
        if (int w = lca(u, v); w == -1) {
            while (v != -1)
                swap(x ^ 1, fae[v]), swap(u, fa[v]), swap(u, v);
        }
        else {
            int z = u, dir = 0; F f = e[x].flow;
            vector<int> cyc = {x};
            for (int d : {0, 1})

```

```

    for (int i = (d ? u : v); i != w; i = fa[i]) {
        cyc.push_back(fae[i] ^ d);
        if (chmin(f, e[fae[i] ^ d].flow)) z = i, dir = d;
    }
    for (int i : cyc) {
        e[i].flow -= f; e[i ^ 1].flow += f;
        cost += f * e[i].cost;
    }
    if (dir) x ^= 1, swap(u, v);
    while (u != z)
        swap(x ^= 1, fae[v]), swap(u, fa[v]), swap(u, v);
}
}
void dfs(int u) {
    vis[u] = visc;
    for (int i : g[u])
        if (int v = e[i].to; vis[v] != visc and e[i].flow)
            fa[v] = u, fae[v] = i, dfs(v);
}
C simplex() {
    fa.assign(g.size(), -1); fae.assign(g.size(), -1);
    C cost = 0; ++visc; dfs(0);
    for (int fail = 0; fail < ssize(e); )
        for (int i = 0; i < ssize(e); i++)
            if (e[i].flow and e[i].cost < phi(e[i ^ 1].to) - phi(e[i].to))
                fail = 0, pushflow(i, cost), ++visc;
            else ++fail;
    return cost;
}
};

```

4.5 Kuhn Munkres [a138de]

```

template <typename T>
struct KM { // 0-based
    const T INF = 1 << 30;
    T w[N][N], hl[N], hr[N], slk[N];
    int fl[N], fr[N], pre[N], n;
    bool vl[N], vr[N];
    queue <int> q;
    KM () {}
    void init(int _n) {
        n = _n;
        for (int i = 0; i < n; ++i)
            for (int j = 0; j < n; ++j) w[i][j] = -INF;
    }
    void add_edge(int a, int b, T wei) { w[a][b] = wei; }
    bool check(int x) {
        if (vl[x] = 1, ~fl[x])
            return q.push(fl[x]), vr[fl[x]] = 1;
        while (~x) swap(x, fr[fl[x] = pre[x]]);
        return 0;
    }
    void bfs(int s) {
        fill(slk, slk + n, INF), fill(vl, vl + n, 0);
        fill(vr, vr + n, 0);
        while (!q.empty()) q.pop();
        q.push(s), vr[s] = 1;
        while (true) {
            T d;
            while (!q.empty()) {
                int y = q.front(); q.pop();
                for (int x = 0; x < n; ++x)
                    if (!vl[x] && slk[x] >= (d = hl[x] + hr[y] - w[x][y]))
                        if (pre[x] = y, d) slk[x] = d;
                        else if (!check(x)) return;
            }
            d = INF;
            for (int x = 0; x < n; ++x)
                if (!vl[x] && d > slk[x]) d = slk[x];
            for (int x = 0; x < n; ++x) {
                if (vl[x]) hl[x] += d;
                else slk[x] -= d;
                if (vr[x]) hr[x] -= d;
            }
            for (int x = 0; x < n; ++x)
                if (!vl[x] && !slk[x] && !check(x)) return;
        }
    }
};

```

```

    }
}
T solve() {
    fill(fl, fl + n, -1), fill(fr, fr + n, -1);
    fill(hr, hr + n, 0);
    for (int i = 0; i < n; ++i)
        hl[i] = *max_element(w[i], w[i] + n);
    for (int i = 0; i < n; ++i) bfs(i);
    T res = 0;
    for (int i = 0; i < n; ++i) res += w[i][fl[i]];
    return res;
}
};

```

4.6 Stoer Wagner (Min-cut) [ac255a]

```

struct SW {
    int g[N][N], sum[N], n;
    bool vis[N], dead[N];
    void init(int _n) {
        n = _n;
        for (int i = 0; i < n; ++i) fill(g[i], g[i] + n, 0);
        fill(dead, dead + n, false);
    }
    void add_edge(int u, int v, int w) {
        g[u][v] += w, g[v][u] += w;
    }
    int run() {
        int ans = 1 << 30;
        for (int round = 0; round + 1 < n; ++round) {
            fill(vis, vis + n, false), fill(sum, sum + n, 0);
            int num = 0, s = -1, t = -1;
            while (num < n - round) {
                int now = -1;
                for (int i = 0; i < n; ++i) if (!vis[i] && !dead[i]) {
                    if (now == -1 || sum[now] < sum[i]) now = i;
                }
                s = t, t = now;
                vis[now] = true, num++;
                for (int i = 0; i < n; ++i) if (!vis[i] && !dead[i]) {
                    dead[i] = true;
                    sum[i] += g[now][i];
                }
            }
            ans = min(ans, sum[t]);
            for (int i = 0; i < n; ++i) {
                g[i][s] += g[i][t];
                g[s][i] += g[t][i];
            }
            dead[t] = true;
        }
        return ans;
    }
};

```

4.7 GomoryHu Tree [90ead2]

```

vector <array <int, 3>> GomoryHu(Dinic <int> flow) {
    // Tree edge min = mincut (0-based)
    int n = flow.n;
    vector <array <int, 3>> ans;
    vector <int> rt(n);
    for (int i = 1; i < n; ++i) {
        int t = rt[i];
        flow.reset();
        ans.pb({i, t, flow.solve(i, t)});
        flow.bfs();
        for (int j = i + 1; j < n; ++j)
            if (rt[j] == t && flow.dis[j] != -1) rt[j] = i;
    }
    return ans;
}

```

4.8 General Graph Matching [2b7f20]

```

struct Matching { // 0-based
    int fa[N], pre[N], match[N], s[N], v[N], n, tk;
    vector <int> g[N];
    queue <int> q;
    int Find(int u) {

```

```

    return u == fa[u] ? u : fa[u] = Find(fa[u]);
}
int lca(int x, int y) {
    tk++;
    x = Find(x), y = Find(y);
    for (; ; swap(x, y)) {
        if (x != n) {
            if (v[x] == tk) return x;
            v[x] = tk;
            x = Find(pre[match[x]]);
        }
    }
}
void blossom(int x, int y, int l) {
    while (Find(x) != l) {
        pre[x] = y, y = match[x];
        if (s[y] == 1) q.push(y), s[y] = 0;
        if (fa[x] == x) fa[x] = l;
        if (fa[y] == y) fa[y] = l;
        x = pre[y];
    }
}
bool bfs(int r) {
    for (int i = 0; i <= n; ++i) fa[i] = i, s[i] = -1;
    while (!q.empty()) q.pop();
    q.push(r);
    s[r] = 0;
    while (!q.empty()) {
        int x = q.front(); q.pop();
        for (int u : g[x]) {
            if (s[u] == -1) {
                pre[u] = x, s[u] = 1;
                if (match[u] == n) {
                    for (int a = u, b = x, last; b != n; a = last, b = pre[a])
                        last = match[b], match[b] = a, match[a] = b;
                    return true;
                }
                q.push(match[u]);
                s[match[u]] = 0;
            } else if (!s[u] && Find(u) != Find(x)) {
                int l = lca(u, x);
                blossom(x, u, l);
                blossom(u, x, l);
            }
        }
    }
    return false;
}
int solve() {
    int res = 0;
    for (int x = 0; x < n; ++x) {
        if (match[x] == n) res += bfs(x);
    }
    return res;
}
void init(int _n) {
    n = _n, tk = 0;
    for (int i = 0; i <= n; ++i) match[i] = pre[i] = n;
    for (int i = 0; i < n; ++i) g[i].clear(), v[i] = 0;
}
void add_edge(int u, int v) {
    g[u].push_back(v), g[v].push_back(u);
}
};

```

4.9 Flow notes

- Bipartite Matching Restore Answer
runBfs(); Answer is $\{vis[x]|x \in L\} \cup \{vis[x]|x \in R\}$
- Bipartite Minimum Weight Vertex Covering
 $S \rightarrow \{x|x \in L\}$, cap = weight of vertex x
 $\{x|x \in L\} \rightarrow \{y|y \in R\}$, cap = ∞
 $\{y|y \in R\} \rightarrow T$, cap = weight of vertex y
 For general version, change Dinic to MCMF and:
 $S \rightarrow \{x|x \in L\}$, cap = weight of vertex x , cost = 0
 $\{x|x \in L\} \rightarrow \{y|y \in R\}$, cap = ∞ , cost = $-w$
 $\{y|y \in R\} \rightarrow T$, cap = weight of vertex y , cost = 0
- Useful Lemma
 (Bipartite Maximum Weight Independent Set) +
 (Bipartite Minimum Weight Vertex Covering) = weight sum

- Min Cut Model
 choose A but not choose B cost $x: A \rightarrow B$, cap = x
 choose A cost $x: A \rightarrow T$, cap = x
 not choose A cost $x: S \rightarrow A$, cap = x
 choose A gain $x \implies$ not choose A cost x , tot+ = x
 choose A and choose B cost x : NO!!!
 Bipartite: can flip one side
- Min Cut Restore Answer
 runBfs(); Answer is $\{vis[x]|x \in V\}$
- Maximum/Minimum flow with lower bound / Circulation problem
 - Construct super source S and sink T .
 - For each edge (x, y, l, u) , connect $x \rightarrow y$ with capacity $u - l$.
 - For each vertex v , denote by $in(v)$ the difference between the sum of incoming lower bounds and the sum of outgoing lower bounds.
 - If $in(v) > 0$, connect $S \rightarrow v$ with capacity $in(v)$, otherwise, connect $v \rightarrow T$ with capacity $-in(v)$.
 - To maximize, connect $t \rightarrow s$ with capacity ∞ (skip this in circulation problem), and let f be the maximum flow from S to T . If $f \neq \sum_{v \in V, in(v) > 0} in(v)$, there's no solution. Otherwise, the maximum flow from s to t is the answer.
 - To minimize, let f be the maximum flow from S to T . Connect $t \rightarrow s$ with capacity ∞ and let the flow from S to T be f' . If $f + f' \neq \sum_{v \in V, in(v) > 0} in(v)$, there's no solution. Otherwise, f' is the answer.
 - The solution of each edge e is $l_e + f_e$, where f_e corresponds to the flow of edge e on the graph.
- Construct minimum vertex cover from maximum matching M on bipartite graph (X, Y)
 - Redirect every edge: $y \rightarrow x$ if $(x, y) \in M$, $x \rightarrow y$ otherwise.
 - DFS from unmatched vertices in X .
 - $x \in X$ is chosen iff x is unvisited.
 - $y \in Y$ is chosen iff y is visited.
- Minimum cost cyclic flow
 - Construct super source S and sink T
 - For each edge (x, y, c) , connect $x \rightarrow y$ with $(cost, cap) = (c, 1)$ if $c > 0$, otherwise connect $y \rightarrow x$ with $(cost, cap) = (-c, 1)$
 - For each edge with $c < 0$, sum these cost as K , then increase $d(y)$ by 1, decrease $d(x)$ by 1
 - For each vertex v with $d(v) > 0$, connect $S \rightarrow v$ with $(cost, cap) = (0, d(v))$
 - For each vertex v with $d(v) < 0$, connect $v \rightarrow T$ with $(cost, cap) = (0, -d(v))$
 - Flow from S to T , the answer is the cost of the flow $C + K$
- Maximum density induced subgraph
 - Binary search on answer, suppose we're checking answer T
 - Construct a max flow model, let K be the sum of all weights
 - Connect source $s \rightarrow v, v \in G$ with capacity K
 - For each edge (u, v, w) in G , connect $u \rightarrow v$ and $v \rightarrow u$ with capacity w
 - For $v \in G$, connect it with sink $v \rightarrow t$ with capacity $K + 2T - (\sum_{e \in E(v)} w(e)) - 2w(v)$
 - T is a valid answer if the maximum flow $f < K|V|$
- Minimum weight edge cover
 - For each $v \in V$ create a copy v' , and connect $u' \rightarrow v'$ with weight $w(u, v)$.
 - Connect $v \rightarrow v'$ with weight $2\mu(v)$, where $\mu(v)$ is the cost of the cheapest edge incident to v .
 - Find the minimum weight perfect matching on G' .
- Project selection problem
 - If $p_v > 0$, create edge (s, v) with capacity p_v ; otherwise, create edge (v, t) with capacity $-p_v$.
 - Create edge (u, v) with capacity w with w being the cost of choosing u without choosing v .
 - The mincut is equivalent to the maximum profit of a subset of projects.

5 String

5.1 AC Automaton [b9fe7c]

```

struct AC {
    int ch[N][26], to[N][26], fail[N], sz;
    vector<int> g[N];
    int cnt[N];
    AC() {sz = 0, extend();}
    void extend() {fill(ch[sz], ch[sz] + 26, 0), sz++;}
    int nxt(int u, int v) {
        if (!ch[u][v]) ch[u][v] = sz, extend();
        return ch[u][v];
    }
    int insert(string s) {
        int now = 0;
        for (char c : s) now = nxt(now, c - 'a');
        cnt[now]++;
    }
};

```

```

    return now;
}
void build_fail() {
    queue<int> q;
    for (int i = 0; i < 26; ++i) if (ch[0][i]) {
        to[0][i] = ch[0][i];
        q.push(ch[0][i]);
        g[0].push_back(ch[0][i]);
    }
    while (!q.empty()) {
        int v = q.front(); q.pop();
        for (int j = 0; j < 26; ++j) {
            to[v][j] = ch[v][j] ? ch[v][j] : to[fail[v]][j];
        }
        for (int i = 0; i < 26; ++i) if (ch[v][i]) {
            int u = ch[v][i], k = fail[v];
            while (k && !ch[k][i]) k = fail[k];
            if (ch[k][i]) k = ch[k][i];
            fail[u] = k;
            cnt[u] += cnt[k], g[k].push_back(u);
            q.push(u);
        }
    }
}
int match(string &s) {
    int now = 0, ans = 0;
    for (char c : s) {
        now = to[now][c - 'a'];
        ans += cnt[now];
    }
    return ans;
}
};

```

5.2 Lyndon Factorization [a9eeb0]

```

// partition s = w[0] + w[1] + ... + w[k-1],
// w[0] >= w[1] >= ... >= w[k-1]
// each w[i] strictly smaller than all its suffix
vector<string> duval(const string &s) {
    vector<string> ans;
    for (int n = (int)s.size(), i = 0, j, k; i < n; ) {
        for (j = i + 1, k = i; j < n && s[k] <= s[j]; j++)
            k = (s[k] < s[j] ? i : k + 1);
        for (; i <= k; i += j - k)
            ans.pb(s.substr(i, j - k)); // s.substr(L, Len)
    }
    return ans;
}

```

5.3 KMP [5ac553]

```

vector<int> build_fail(string &s) {
    vector<int> f(s.length() + 1, 0);
    int k = 0;
    for (int i = 1; i < s.length(); ++i) {
        while (k && s[k] != s[i])
            k = f[k];
        if (s[k] == s[i])
            k++;
        f[i + 1] = k;
    }
    return f;
}
int match(string &s, string &t) {
    vector<int> f = build_fail(t);
    int k = 0, ans = 0;
    for (int i = 0; i < s.length(); ++i) {
        while (k && s[i] != t[k])
            k = f[k];
        if (s[i] == t[k])
            k++;
        if (k == t.length())
            ans++, k = f[k];
    }
    return ans;
}

```

5.4 Manacher [643e55]

```
int z[MAXN]; //0-base
```

```

/* center i: radius z[i * 2 + 1] / 2
   center i, i + 1: radius z[i * 2 + 2] / 2
   both aba, abba have radius 2 */
void Manacher(string tmp) {
    string s = "%";
    int l = 0, r = 0;
    for (char c : tmp) s.pb(c), s.pb('%');
    for (int i = 0; i < s.size(); ++i) {
        z[i] = r > i ? min(z[2 * l - i], r - i) : 1;
        while (i - z[i] >= 0 && i + z[i] < s.size()
            && s[i + z[i]] == s[i - z[i]]) ++z[i];
        if (z[i] + i > r) r = z[i] + i, l = i;
    }
}

```

5.5 Minimum Rotate [acab8e]

```

string mcp(string s) {
    int n = s.size(), i = 0, j = 1;
    s += s;
    while (i < n && j < n) {
        int k = 0;
        while (k < n && s[i + k] == s[j + k]) k++;
        if (s[i + k] <= s[j + k]) j += k + 1;
        else i += k + 1;
        if (i == j) j++;
    }
    int ans = (i < n ? i : j);
    return s.substr(ans, n);
}

```

5.6 Palindrome Tree [1f78bf]

```

struct PalindromicTree {
    struct node {
        int nxt[26], fail, len; // num = depth of fail link
        int cnt, num; // cnt = occur, num = #pal_suffix of
            this node
        node(int l = 0) : nxt{}, fail(0), len(l), cnt(0), num
            (0) {}
    };
    vector<node> st; vector<int> s; int last, n;
    void init() {
        st.clear(); s.clear(); last = 1; n = 0;
        st.pb(0); st.pb(-1);
        st[0].fail = 1; s.pb(-1);
    }
    int getFail(int x) {
        while (s[n - st[x].len - 1] != s[n]) x = st[x].fail;
        return x;
    }
    void add(int c) {
        s.pb(c - 'a'); ++n;
        int cur = getFail(last);
        if (!st[cur].nxt[c]) {
            int now = sz(st);
            st.pb(st[cur].len + 2);
            st[now].fail = st[getFail(st[cur].fail)].nxt[c];
            st[cur].nxt[c] = now;
            st[now].num = st[st[now].fail].num + 1;
        }
        last = st[cur].nxt[c]; ++st[last].cnt;
    }
    void dpCnt() {
        for (int i = sz(st) - 1; i >= 0; i--) {
            auto nd = st[i];
            st[nd.fail].cnt += nd.cnt;
        }
    }
    int size() { return (int)st.size() - 2; }
};

```

5.7 Palindrome Partition [c85c05]

```

// in PAM
/* node */ int dif = 0, slink = 0, g = 0;
vector<int> dp = {0};
// add
if (!st[cur].nxt[c]) {
    // ...
    st[now].dif = st[now].len - st[st[now].fail].len;
}

```

```

if (st[now].dif == st[st[now].fail].dif)
    st[now].slink = st[st[now].fail].slink;
else st[now].slink = st[now].fail;
}
dp.pb(0);
for (int x = last; x > 1; x = st[x].slink) {
    st[x].g = dp[n - st[st[x].slink].len - st[x].dif];
    if (st[x].dif == st[st[x].fail].dif)
        st[x].g = min(st[x].g, st[st[x].fail].g);
    dp[n] = min(dp[n], st[x].g + 1);
}

```

5.8 Repetition [f3da14]

```

int to_left[N], to_right[N];
vector<array<int, 3>> rep; // L, r, Len.
// substr( [l, r], len * 2) are tandem
void findRep(string &s, int l, int r) {
    if (r - l == 1) return;
    int m = l + r >> 1;
    findRep(s, l, m), findRep(s, m, r);
    string sl = s.substr(l, m - l);
    string sr = s.substr(m, r - m);
    vector<int> Z = buildZ(sr + "#" + sl);
    for (int i = l; i < m; ++i)
        to_right[i] = Z[r - m + 1 + i - l];
    reverse(all(sl));
    Z = buildZ(sl);
    for (int i = l; i < m; ++i)
        to_left[i] = Z[m - i - l];
    reverse(all(sl));
    for (int i = l; i + 1 < m; ++i) {
        int k1 = to_left[i], k2 = to_right[i + 1];
        int len = m - i - 1;
        if (k1 < 1 || k2 < 1 || len < 2) continue;
        int tl = max(1, len - k2), tr = min(len - 1, k1);
        if (tl <= tr) rep.pb({i + 1 - tr, i + 1 - tl, len});
    }
    Z = buildZ(sr);
    for (int i = m; i < r; ++i) to_right[i] = Z[i - m];
    reverse(all(sl)), reverse(all(sr));
    Z = buildZ(sl + "#" + sr);
    for (int i = m; i < r; ++i)
        to_left[i] = Z[m - l + 1 + r - i - 1];
    reverse(all(sl)), reverse(all(sr));
    for (int i = m; i + 1 < r; ++i) {
        int k1 = to_left[i], k2 = to_right[i + 1];
        int len = i - m + 1;
        if (k1 < 1 || k2 < 1 || len < 2) continue;
        int tl = max(len - k2, 1), tr = min(len - 1, k1);
        if (tl <= tr)
            rep.pb({i + 1 - len - tr, i + 1 - len - tl, len});
    }
    Z = buildZ(sr + "#" + sl);
    for (int i = l; i < m; ++i)
        if (Z[r - m + 1 + i - l] >= m - i)
            rep.pb({i, i, m - i});
}

```

5.9 Suffix Array [5fde59]

```

#define FOR(i, a, b) for(int i = a; i <= b; i++)
#define ROF(i, a, b) for(int i = a; i >= b; i--)
template<int N>
struct SA {
    int sa[N], tmp[2][N], c[N], rk[N], lcp[N];
    void buildSA(const string &s) { // 9041ad
        int *x = tmp[0], *y = tmp[1], n = sz(s), m = 256;
        FOR(i, 0, m - 1) c[i] = 0;
        FOR(i, 0, n - 1) c[x[i]] = s[i]++;
        FOR(i, 1, m - 1) c[i] += c[i - 1];
        ROF(i, n - 1, 0) sa[--c[x[i]]] = i;
        for (int k = 1; k < n; k <= 1) {
            FOR(i, 0, m - 1) c[i] = 0;
            FOR(i, 0, n - 1) c[x[i]]++;
            FOR(i, 1, m - 1) c[i] += c[i - 1];
            int p = 0;
            FOR(i, n - k, n - 1) y[p++] = i;
            FOR(i, 0, n - 1) if (sa[i] >= k) y[p++] = sa[i] - k;
            ROF(i, n - 1, 0) sa[--c[x[y[i]]]] = y[i];
            y[sa[0]] = p = 0;

```

```

        FOR(i, 1, n - 1) {
            int a = sa[i], b = sa[i - 1];
            if (!(x[a] == x[b] && a + k < n && b + k < n &&
                x[a + k] == x[b + k])) p++;
            y[sa[i]] = p;
        }
        if (n == p + 1) break;
        swap(x, y), m = p + 1;
    }
} // sa[rk[i]] = i
void buildLCP(const string &s) { // 26131c
    int n = sz(s), val = 0;
    FOR(i, 0, n - 1) rk[sa[i]] = i;
    FOR(i, 0, n - 1) {
        if (!rk[i]) lcp[rk[i]] = 0;
        else {
            if (val) val--;
            int p = sa[rk[i] - 1];
            while (val + i < n && val + p < n && s[val + i]
                == s[val + p]) val++;
            lcp[rk[i]] = val;
        }
    }
} // lcp[i] = LCP(sa[i - 1], sa[i])
void build(const string &s) {
    buildSA(s);
    buildLCP(s);
}
};

```

5.10 SAIS (C++20) [b2c106]

```

auto sais(const auto &s) { // afa632
    const int n = sz(s), z = ranges::max(s) + 1;
    if (n == 1) return vector{0};
    vector<int> c(z); for (int x : s) ++c[x];
    partial_sum(iter(c), begin(c));
    vector<int> sa(n); auto I = views::iota(0, n);
    vector<bool> t(n, true);
    for (int i = n - 2; i >= 0; --i)
        t[i] = (s[i] == s[i + 1] ? t[i + 1] : s[i] < s[i + 1]);
    auto is_lms = views::filter([&t](int x) {
        return x && t[x] && !t[x - 1];
    });
    auto induce = [&] {
        for (auto x = c; int y : sa)
            if (y-- if (!t[y]) sa[x[s[y] - 1]++] = y;
        for (auto x = c; int y : sa | views::reverse)
            if (y-- if (t[y]) sa[--x[s[y]]] = y;
    };
    vector<int> lms, q(n); lms.reserve(n);
    for (auto x = c; int i : I | is_lms)
        q[i] = sz(lms), lms.push_back(sa[--x[s[i]]] = i);
    induce(); vector<int> ns(sz(lms));
    for (int j = -1, nz = 0; int i : sa | is_lms) {
        if (j >= 0) {
            int len = min({n - i, n - j, lms[q[i] + 1] - i});
            ns[q[i]] = nz += lexicographical_compare(
                begin(s) + j, begin(s) + j + len,
                begin(s) + i, begin(s) + i + len);
        }
        j = i;
    }
    fill(iter(sa), 0); auto nsa = sais(ns);
    for (auto x = c; int y : nsa | views::reverse)
        y = lms[y], sa[--x[s[y]]] = y;
    return induce(), sa;
}

struct Suffix {
    int n; vector<int> sa, lcp, ra;
    Suffix(const auto &s, int _n) : n(_n), lcp(n), ra(n) {
        // 86c942
        vector<int> s(n + 1); // s[n] = 0;
        copy_n(s.begin(), n, begin(s)); // _s shouldn't
            contain 0
        sa = sais(s); sa.erase(sa.begin());
        for (int i = 0; i < n; ++i) ra[sa[i]] = i;
        for (int i = 0, h = 0; i < n; ++i) {
            if (!ra[i]) { h = 0; continue; }
            for (int j = sa[ra[i] - 1]; max(i, j) + h < n &&
                s[i + h] == s[j + h];) ++h;

```

```

    lcp[ra[i]] = h ? h-- : 0;
}
} // lcp[i] = LCP(sa[i - 1], sa[i])
};

```

5.11 Suffix Automaton [277d1d]

```

struct SAM {
    int ch[N][26], len[N], link[N], pos[N], cnt[N], sz;
    // node -> strings with the same endpos set
    // length in range [len(link) + 1, len]
    // node's endpos set -> pos in the subtree of node
    // link -> longest suffix with different endpos set
    // len -> longest suffix
    // pos -> end position
    // cnt -> size of endpos set
    SAM () {len[0] = 0, link[0] = -1, pos[0] = 0, cnt[0]
        = 0, sz = 1;}
    void build(string s) {
        int last = 0;
        for (int i = 0; i < s.length(); ++i) {
            char c = s[i];
            int cur = sz++;
            len[cur] = len[last] + 1, pos[cur] = i + 1;
            int p = last;
            while (~p && !ch[p][c - 'a'])
                ch[p][c - 'a'] = cur, p = link[p];
            if (p == -1) link[cur] = 0;
            else {
                int q = ch[p][c - 'a'];
                if (len[p] + 1 == len[q]) {
                    link[cur] = q;
                } else {
                    int nxt = sz++;
                    len[nxt] = len[p] + 1, link[nxt] = link[q];
                    pos[nxt] = pos[q];
                    for (int j = 0; j < 26; ++j)
                        ch[nxt][j] = ch[q][j];
                    while (~p && ch[p][c - 'a'] == q)
                        ch[p][c - 'a'] = nxt, p = link[p];
                    link[q] = link[cur] = nxt;
                }
            }
            cnt[cur]++;
            last = cur;
        }
        vector<int> p(sz);
        iota(all(p), 0);
        sort(all(p),
            [&](int i, int j) {return len[i] > len[j];});
        for (int i = 0; i < sz; ++i)
            cnt[link[p[i]]] += cnt[p[i]];
    }
} sam;

```

5.12 Z Value [a8c33c]

```

vector<int> build(string s) {
    int n = s.length();
    vector<int> Z(n);
    int l = 0, r = 0;
    for (int i = 0; i < n; ++i) {
        Z[i] = max(min(Z[i - 1], r - i), 0);
        while (i + Z[i] < s.size() && s[Z[i]] == s[i + Z[i]]) {
            l = i, r = i + Z[i], Z[i]++;
        }
    }
    return Z;
}

```

6 Math

6.1 Characteristic Polynomial [eabd3c]

```

#define rep(x, y, z) for (int x=y; x<z; x++)
using vi = vector<int>; using vvi = vector<vi>;
void Hessenberg(vvi &H, int N) { // 0ddcb8
    for (int i = 0; i < N - 2; ++i) {
        for (int j = i + 1; j < N; ++j) if (H[j][i]) {
            rep(k, i, N) swap(H[i+1][k], H[j][k]);

```

```

            rep(k, 0, N) swap(H[k][i+1], H[k][j]);
            break;
        }
        if (!H[i + 1][i]) continue;
        for (int j = i + 2; j < N; ++j) {
            int co = mul(modinv(H[i + 1][i]), H[j][i]);
            rep(k, i, N) subeq(H[j][k], mul(H[i+1][k], co));
            rep(k, 0, N) addeq(H[k][i+1], mul(H[k][j], co));
        }
    }
}
vi CharacteristicPoly(vvi A) { // 5213a6
    int N = (int)A.size(); Hessenberg(A, N);
    vvi P(N + 1, vi(N + 1)); P[0][0] = 1;
    for (int i = 1; i <= N; ++i) {
        rep(j, 0, i+1) P[i][j] = j ? P[i-1][j-1] : 0;
        for (int j = i - 1, val = 1; j >= 0; --j) {
            int co = mul(val, A[j][i - 1]);
            rep(k, 0, j+1) subeq(P[i][k], mul(P[j][k], co));
            if (j) val = mul(val, A[j][j - 1]);
        }
    }
    if (N & 1) for (int &x: P[N]) x = sub(0, x);
    return P[N]; // test: 2021 PTZ Korea K
}

```

6.2 Discrete Logarithm [4c39b3]

```

int BSGS(int s, int x, int y, int m) { // 247dc4
    constexpr int kStep = 32000;
    unordered_map<int, int> p;
    int b = 1;
    for (int i = 0; i < kStep; ++i) {
        p[y] = i;
        y = 1ll * y * x % m;
        b = 1ll * b * x % m;
    }
    for (int i = 0; i < m + 10; i += kStep) {
        s = 1ll * s * b % m;
        if (p.find(s) != p.end()) return i + kStep - p[s];
    }
    return -1; // returns: j where s*(x^j) = y (mod m)
}
int DiscreteLog(int x, int y, int m) { // 579300
    if (m == 1) return 0;
    int s = 1;
    for (int i = 0; i < 100; ++i) {
        if (s == y) return i;
        s = 1ll * s * x % m;
    }
    if (s == y) return 100;
    int p = 100 + BSGS(s, x, y, m);
    if (fpow(x, p, m) != y) return -1;
    return p; //returns: x^p = y (mod m)
}

```

6.3 Extgcd [c51ae9]

```

// ax+ny = 1, ax+ny == ax == 1 (mod n)
void extgcd(ll x, ll y, ll &g, ll &a, ll &b) {
    if (y == 0) g = x, a = 1, b = 0;
    else extgcd(y, x % y, g, b, a), b -= (x / y) * a;
}

```

6.4 Floor Sum [49de67]

```

// sum^{n-1}_0 floor((a * i + b) / m) in Log(n + m + a + b)
ll floor_sum(ll n, ll m, ll a, ll b) {
    ll ans = 0;
    if (a >= m) ans += (n - 1) * n * (a / m) / 2, a %= m;
    if (b >= m) ans += n * (b / m), b %= m;
    ll y_max = (a * n + b) / m, x_max = (y_max * m - b);
    if (y_max == 0) return ans;
    ans += (n - (x_max + a - 1) / a) * y_max;
    ans += floor_sum(y_max, a, m, (a - x_max % a) % a);
    return ans;
}

```

6.5 Factorial Mod P^k [c324f3]

```
// O(p^k + Log^2 n), pk = p^k
ll prod[MAXP];
ll fac_no_p(ll n, ll p, ll pk) {
    prod[0] = 1;
    for (int i = 1; i <= pk; ++i)
        if (i % p) prod[i] = prod[i - 1] * i % pk;
        else prod[i] = prod[i - 1];
    ll rt = 1;
    for (; n; n /= p) {
        rt = rt * mpow(prod[pk], n / pk, pk) % pk;
        rt = rt * prod[n % pk] % pk;
    }
    return rt;
} // (n! without factor p) % p^k
```

6.6 Gaussian Elimination [0465e4]

```
using vi = vector<int>; // be careful if A.empty()
using vvi = vector<vi>; // ensure that 0 <= x < mod
pair<vi, vvi> gauss(vvi A, vi b) { // solve Ax=b
    const int N = (int)A.size(), M = (int)A[0].size();
    vector<int> depv, free(M, true); int rk = 0;
    for (int i = 0; i < M; i++) {
        int p = -1;
        for (int j = rk; j < N; j++)
            if (p == -1 || abs(A[j][i]) > abs(A[p][i]))
                p = j;
        if (p == -1 || A[p][i] == 0) continue;
        swap(A[p], A[rk]); swap(b[p], b[rk]);
        const int inv = modinv(A[rk][i]);
        for (int &x : A[rk]) x = mul(x, inv);
        b[rk] = mul(b[rk], inv);
        for (int j = 0; j < N; j++) if (j != rk) {
            int z = A[j][i];
            for (int k = 0; k < M; k++)
                A[j][k] = sub(A[j][k], mul(z, A[rk][k]));
            b[j] = sub(b[j], mul(z, b[rk]));
        }
        depv.push_back(i); free[i] = false; ++rk;
    }
    for (int i = rk; i < N; i++)
        if (b[i] != 0) return {{}, {}}; // not consistent
    vi x(M); vvi h;
    for (int i = 0; i < rk; i++) x[depv[i]] = b[i];
    for (int i = 0; i < M; i++) if (free[i]) {
        h.emplace_back(M); h.back()[i] = 1;
        for (int j = 0; j < rk; j++)
            h.back()[depv[j]] = sub(0, A[j][i]);
    }
    return {x, h}; // solution = x + span(h[i])
}
```

6.7 Linear Function Mod Min [93cc4b]

```
ll topos(ll x, ll m) {x %= m; if (x < 0) x += m; return x;}
//min value of ax + b (mod m) for x \in [0, n - 1]. O(Log m)
ll min_rem(ll n, ll m, ll a, ll b) { // 578b62
    a = topos(a, m), b = topos(b, m);
    for (ll g = __gcd(a, m); g > 1; ) return g * min_rem(n / g, m / g, a / g, b / g) + (b % g);
    for (ll nn, nm, na, nb; a; n = nn, m = nm, a = na, b = nb) {
        if (a <= m - a) {
            nn = (a * (n - 1) + b) / m;
            if (!nn) break;
            nn += (b < a);
            nm = a, na = topos(-m, a);
            nb = b < a ? b : topos(b - m, a);
        } else {
            ll lst = b - (n - 1) * (m - a);
            if (lst >= 0) {b = lst; break;}
            nn = -(lst / m) + (lst % m < -a) + 1;
            nm = m - a, na = m % (m - a), nb = b % (m - a);
        }
    }
    return b;
}
//min value of ax + b (mod m) for x \in [0, n - 1],
//also return min x to get the value. O(Log m)
//{value, x}
```

```
pair<ll, ll> min_rem_pos(ll n, ll m, ll a, ll b) { //
    fd6550
    a = topos(a, m), b = topos(b, m);
    ll mn = min_rem(n, m, a, b), g, ia, ib;
    extgcd(a, m, g, ia, ib);
    //ax = (mn - b) (mod m)
    ll x = (ia + m) * ((mn - b + m) / g) % (m / g);
    return {mn, x};
}
```

6.8 MillerRabin PollardRho [9d0778]

```
// n < 4,759,123,141      3 : 2, 7, 61
// n < 1,122,004,669,633  4 : 2, 13, 23, 1662803
// n < 3,474,749,660,383  6 : primes <= 13
// n < 2^64              7 :
// 2, 325, 9375, 28178, 450775, 9780504, 1795265022
ll mul(ll a, ll b, ll n) { return (__int128)a * b % n; }
vector<ll> chk = {2, 325, 9375, 28178, 450775, 9780504, 1795265022};
ll Pow(ll a, ll b, ll n) {ll res = 1; for (; b; b >>= 1, a = mul(a, a, n)) if (b & 1) res = mul(res, a, n); return res;}
bool check(ll a, ll d, int s, ll n) { // 874cd4
    a = Pow(a, d, n);
    if (a <= 1) return 1;
    for (int i = 0; i < s; ++i, a = mul(a, a, n)) {
        if (a == 1) return 0;
        if (a == n - 1) return 1;
    }
    return 0;
}
bool IsPrime(ll n) { // b8bfee
    if (n < 2) return 0;
    if (n % 2 == 0) return n == 2;
    ll d = n - 1, s = 0;
    while (d % 2 == 0) d >>= 1, ++s;
    for (ll i : chk) if (!check(i, d, s, n)) return 0;
    return 1;
}
const vector<ll> small = {2, 3, 5, 7, 11, 13, 17, 19};
ll FindFactor(ll n) { // 697b88
    if (IsPrime(n)) return 1;
    for (ll p : small) if (n % p == 0) return p;
    ll x, y = 2, d, t = 1;
    auto f = [&](ll a) {return (mul(a, a, n) + t) % n;};
    for (int l = 2; ; l <= 1) {
        x = y;
        int m = min(1, 32);
        for (int i = 0; i < l; i += m) {
            d = 1;
            for (int j = 0; j < m; ++j) {
                y = f(y), d = mul(d, abs(x - y), n);
            }
            ll g = __gcd(d, n);
            if (g == n) {
                l = 1, y = 2, ++t;
                break;
            }
            if (g != 1) return g;
        }
    }
}
map<ll, int> PollardRho(ll n) { // 1cf996
    map<ll, int> res;
    if (n == 1) return res;
    if (IsPrime(n)) return ++res[n], res;
    ll d = FindFactor(n);
    res = PollardRho(n / d);
    auto res2 = PollardRho(d);
    for (auto [x, y] : res2) res[x] += y;
    return res;
}
```

6.9 Lucy Hedgehog [435ed1]

```
ll r, rt, ans = 0; // Goal: S(i, k) = sum_{j < i, lpf(j) > p_k} f(j), S(i) = S(i, inf)
// Find g s.t. S(i, k) = S(i, k-1) - g(k) (S(i / p_k, k - 1) - S(p_k - 1, k-1))
```

```
vector<ll> small, large; // small[i] : S(i) for i = 1
... r^(1/2), large[i] : S(r/i) for i = 1 ... r
^(1/2)
ll get(ll x){ return x <= rt ? small[x] : large[r / x];
}
ll init(ll x){ return x * (x+1) / 2 - 1; }
void lcy(){
    rt = ll(sqrt(r));
    small.resize(rt + 1); large.resize(rt + 1);
    for(int i=1;i<=rt;i++){
        small[i] = init(i); large[i] = init(r / i);
    }
    for(ll k=2;k<=rt;k++){
        if(small[k] == small[k-1]) continue;
        for(ll i = 1; i <= rt && i * k * k <= r; i++){
            large[i] -= k * (get(r / i / k) - small[k-1]);
        }
        for(ll i = rt; i >= k * k; i--){
            small[i] -= k * (small[i/k] - small[k-1]);
        }
    }
}
```

6.10 Quadratic Residue [ac688c]

```
int Jacobi(int a, int m) { // 00fa11
    int s = 1;
    for (; m > 1; ) {
        a %= m;
        if (a == 0) return 0;
        const int r = __builtin_ctz(a);
        if ((r & 1) && ((m + 2) & 4)) s = -s;
        a >>= r;
        if (a & m & 2) s = -s;
        swap(a, m);
    }
    return s;
}
int QuadraticResidue(int a, int p) { // 1a2d08
    if (p == 2) return a & 1;
    const int jc = Jacobi(a, p);
    if (jc == 0) return 0;
    if (jc == -1) return -1;
    int b, d;
    for (; ; ) {
        b = rand() % p;
        d = (11l * b * b + p - a) % p;
        if (Jacobi(d, p) == -1) break;
    }
    int f0 = b, f1 = 1, g0 = 1, g1 = 0, tmp;
    for (int e = (11l + p) >> 1; e; e >>= 1) {
        if (e & 1) {
            tmp = (11l * g0 * f0 + 11l * d * (11l * g1 * f1 %
                p)) % p;
            g1 = (11l * g0 * f1 + 11l * g1 * f0) % p;
            g0 = tmp;
        }
        tmp = (11l * f0 * f0 + 11l * d * (11l * f1 * f1 % p
            )) % p;
        f1 = (21l * f0 * f1) % p;
        f0 = tmp;
    }
    return g0;
}
```

6.11 Simplex [ece29e]

```
struct Simplex { // 0-based
    using T = long double;
    static const int N = 410, M = 30010;
    const T eps = 1e-7;
    int n, m;
    int Left[M], Down[N];
    // Ax <= b, max c^T x. result : v, xi = sol[i]. 1-
    based
    T a[M][N], b[M], c[N], v, sol[N];
    bool eq(T a, T b) {return fabs(a - b) < eps;}
    bool ls(T a, T b) {return a < b && !eq(a, b);}
    void init(int _n, int _m) { // a752a0
        n = _n, m = _m, v = 0;
        for (int i = 0; i < m; ++i) for (int j = 0; j < n;
            ++j) a[i][j] = 0;
```

```
for (int i = 0; i < m; ++i) b[i] = 0;
for (int i = 0; i < n; ++i) c[i] = sol[i] = 0;
}
void pivot(int x, int y) { // 92faa6
    swap(Left[x], Down[y]);
    T k = a[x][y]; a[x][y] = 1;
    vector<int> nz;
    for (int i = 0; i < n; ++i) {
        a[x][i] /= k;
        if (!eq(a[x][i], 0)) nz.push_back(i);
    }
    b[x] /= k;
    for (int i = 0; i < m; ++i) {
        if (i == x || eq(a[i][y], 0)) continue;
        k = a[i][y], a[i][y] = 0;
        b[i] -= k * b[x];
        for (int j : nz) a[i][j] -= k * a[x][j];
    }
    if (eq(c[y], 0)) return;
    k = c[y], c[y] = 0, v += k * b[x];
    for (int i : nz) c[i] -= k * a[x][i];
}
// 0: found solution, 1: no feasible solution, 2:
unbounded
int solve() { // 9f4550
    for (int i = 0; i < n; ++i) Down[i] = i;
    for (int i = 0; i < m; ++i) Left[i] = n + i;
    while (1) {
        int x = -1, y = -1;
        for (int i = 0; i < m; ++i) if (ls(b[i], 0) && (x
            == -1 || b[i] < b[x])) x = i;
        if (x == -1) break;
        for (int i = 0; i < n; ++i) if (ls(a[x][i], 0) &&
            (y == -1 || a[x][i] < a[x][y])) y = i;
        if (y == -1) return 1;
        pivot(x, y);
    }
    while (1) {
        int x = -1, y = -1;
        for (int i = 0; i < n; ++i) if (ls(0, c[i]) && (y
            == -1 || c[i] > c[y])) y = i;
        if (y == -1) break;
        for (int i = 0; i < m; ++i) if (ls(0, a[i][y]) &&
            (x == -1 || b[i] / a[i][y] < b[x] / a[x][y]))
            x = i;
        if (x == -1) return 2;
        pivot(x, y);
    }
    for (int i = 0; i < m; ++i) if (Left[i] < n) sol[
        Left[i]] = b[i];
    return 0;
}
};
```

6.12 Linear Programming Construction

Standard form: maximize $c^T x$ subject to $Ax \leq b$ and $x \geq 0$.
 Dual LP: minimize $b^T y$ subject to $A^T y \geq c$ and $y \geq 0$.
 $\bar{x}$ and $\bar{y}$ are optimal if and only if for all $i \in [1, n]$, either $\bar{x}_i = 0$ or $\sum_{j=1}^m A_{ji} \bar{y}_j = c_i$ holds and for all $i \in [1, m]$ either $\bar{y}_i = 0$ or $\sum_{j=1}^n A_{ij} \bar{x}_j = b_j$ holds.

1. In case of minimization, let $c'_i = -c_i$
2. $\sum_{1 \leq i \leq n} A_{ji} x_i \geq b_j \rightarrow \sum_{1 \leq i \leq n} -A_{ji} x_i \leq -b_j$
3. $\sum_{1 \leq i \leq n} A_{ji} x_i = b_j$
 - $\sum_{1 \leq i \leq n} A_{ji} x_i \leq b_j$
 - $\sum_{1 \leq i \leq n} A_{ji} x_i \geq b_j$
4. If x_i has no lower bound, replace x_i with $x_i - x'_i$

6.13 Estimation

- Number of partitions of n :

n	2	3	4	5	6	7	8	9	20	30	40	50	100
$p(n)$	2	3	5	7	11	15	22	30	627	5604	4e4	2e5	2e8
- Maximum number of divisors:

n	100	1e3	1e6	1e9	1e12	1e15	1e18
$d(i)$	12	32	240	1344	6720	26880	103680
arg	60	840	720720	735134400	963761198400	866421317361600	89761248478661
- $\frac{n}{(2^n)}$

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$\frac{n}{(2^n)}$	2	6	20	70	252	924	3432	12870	48620	184756	7e5	2e6	1e7	4e7	1.5e8
- Number of ways to partition a set of n labeled elements:

n	2	3	4	5	6	7	8	9	10	11	12	13
B_n	2	5	15	52	203	877	4140	21147	115975	7e5	4e6	3e7

6.14 Theorem

Kirchhoff's Theorem

Denote L be a $n \times n$ matrix as the Laplacian matrix of graph G , where $L_{ii} = d(i)$, $L_{ij} = -c$ where c is the number of edge (i, j) in G .

- The number of undirected spanning in G is $|\det(\tilde{L}_{11})|$.
- The number of directed spanning tree rooted at r in G is $|\det(\tilde{L}_{rr})|$.

Tutte's Matrix

Let D be a $n \times n$ matrix, where $d_{ij} = x_{ij}$ (x_{ij} is chosen uniformly at random) if $i < j$ and $(i, j) \in E$, otherwise $d_{ij} = -d_{ji}$. $\frac{rank(D)}{2}$ is the maximum matching on G .

Cayley's Formula

- Given a degree sequence $d_1, d_2, \dots, d_n$ for each *labeled* vertices, there are

$$\frac{(n-2)!}{(d_1-1)!(d_2-1)! \cdots (d_n-1)!}$$

spanning trees.

- Let $T_{n,k}$ be the number of *labeled* forests on n vertices with k components, such that vertex $1, 2, \dots, k$ belong to different components. Then $T_{n,k} = kn^{n-k-1}$.

Erdős-Gallai Theorem

A sequence of non-negative integers $d_1 \geq d_2 \geq \dots \geq d_n$ can be represented as the degree sequence of a finite simple graph on n vertices if and only if $d_1 + d_2 + \dots + d_n$ is even and

$$\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min(d_i, k)$$

holds for all $1 \leq k \leq n$.

Burnside's Lemma

Let X be a set and G be a group that acts on X . For $g \in G$, denote by X^g the elements fixed by g :

$$X^g = \{x \in X \mid gx = x\}$$

Then

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|.$$

Gale-Ryser theorem

A pair of sequences of nonnegative integers $a_1 \geq \dots \geq a_n$ and $b_1, \dots, b_n$ is bigraphic if and only if $\sum_{i=1}^n a_i = \sum_{i=1}^n b_i$ and $\sum_{i=1}^k a_i \leq \sum_{i=1}^n \min(b_i, k)$ holds for every $1 \leq k \leq n$.

Fulkerson-Chen-Anstee theorem

A sequence $(a_1, b_1), \dots, (a_n, b_n)$ of nonnegative integer pairs with $a_1 \geq \dots \geq a_n$ is digraphic if and only if $\sum_{i=1}^n a_i = \sum_{i=1}^n b_i$ and $\sum_{i=1}^k a_i \leq \sum_{i=1}^n \min(b_i, k)$ holds for every $1 \leq k \leq n$.

$$\sum_{i=1}^k \min(b_i, k-1) + \sum_{i=k+1}^n \min(b_i, k) \text{ holds for every } 1 \leq k \leq n.$$

Möbius inversion formula

- $f(n) = \sum_{d|n} g(d) \Leftrightarrow g(n) = \sum_{d|n} \mu(d) f(\frac{n}{d})$
- $f(n) = \sum_{n|d} g(d) \Leftrightarrow g(n) = \sum_{n|d} \mu(\frac{d}{n}) f(d)$

Spherical cap

- A portion of a sphere cut off by a plane.
- r : sphere radius, a : radius of the base of the cap, h : height of the cap, θ : $\arcsin(a/r)$.
- Volume $= \pi h^2(3r-h)/3 = \pi h(3a^2+h^2)/6 = \pi r^3(2+\cos\theta)(1-\cos\theta)^2/3$.
- Area $= 2\pi rh = \pi(a^2+h^2) = 2\pi r^2(1-\cos\theta)$.

Chinese Remainder Theorem

- $x \equiv a_i \pmod{m_i}$
- $M = \prod m_i, M_i = M/m_i$
- $t_i M_i \equiv 1 \pmod{m_i}$
- $x = \sum a_i t_i M_i \pmod{M}$

- Green Hackenbush: Run BCC(bridge), each BCC has Grundy value $g_v = |E| \bmod 2$ (the Grundy value remains unchanged if we identify a circuit as a node). Run DFS with root *ground* and $g_v \leftarrow g_v \oplus (g_h + 1)$ for vertex v and its child h .

6.15 General Purpose Numbers

Bernoulli numbers

$$B_0 = 1, B_1^\pm = \pm \frac{1}{2}, B_2 = \frac{1}{6}, B_3 = 0$$

$$\sum_{j=0}^m \binom{m+1}{j} B_j = 0, \text{EGF is } B(x) = \frac{x}{e^x - 1} = \sum_{n=0}^{\infty} B_n \frac{x^n}{n!}.$$

$$S_m(n) = \sum_{k=1}^n k^m = \frac{1}{m+1} \sum_{k=0}^m \binom{m+1}{k} B_k^+ n^{m+1-k}$$

- Stirling numbers of the second kind Partitions of n distinct elements into exactly k groups.

$$S(n, k) = S(n-1, k-1) + kS(n-1, k), S(n, 1) = S(n, n) = 1$$

$$S(n, k) = \frac{1}{k!} \sum_{i=0}^k (-1)^{k-i} \binom{k}{i} i^n$$

$$x^n = \sum_{i=0}^n S(n, i) (x)_i$$

Pentagonal number theorem

$$\prod_{n=1}^{\infty} (1 - x^n) = 1 + \sum_{k=1}^{\infty} (-1)^k \left(x^{k(3k+1)/2} + x^{k(3k-1)/2} \right)$$

Catalan numbers

$$C_n^{(k)} = \frac{1}{(k-1)n+1} \binom{kn}{n}$$

$$C^{(k)}(x) = 1 + x[C^{(k)}(x)]^k$$

Eulerian numbers

Number of permutations $\pi \in S_n$ in which exactly k elements are greater than the previous element. k : j 's s.t. $\pi(j) > \pi(j+1)$, $k+1$: j 's s.t. $\pi(j) \geq j$, k : j 's s.t. $\pi(j) > j$.

$$E(n, k) = (n-k)E(n-1, k-1) + (k+1)E(n-1, k)$$

$$E(n, 0) = E(n, n-1) = 1$$

$$E(n, k) = \sum_{j=0}^k (-1)^j \binom{n+1}{j} (k+1-j)^n$$

6.16 Integral

$\frac{d}{dx}$	$\sec^2 x$	$-\csc^2 x$	$\tan x \sec x$	$-\cot x \csc x$
$f(x)$	$\tan x$	$\cot x$	$\sec x$	$\csc x$
$\int$	$\ln \sec x $	$-\ln \csc x $	$\ln \tan x + \sec x $	$-\ln \cot x + \csc x $
$\frac{d}{dx}$	$\frac{1}{\sqrt{1-x^2}}$	$\frac{-1}{\sqrt{1-x^2}}$	$\frac{1}{x^2+1}$	$\sec^3 x$
$f(x)$	$\arcsin x$	$\arccos x$	$\arctan x$	$\frac{1}{2} (\sec x \tan x + \ln \sec x + \tan x)$

$$\int_0^{\frac{\pi}{2}} \sin^{2k} x \, dx = \frac{\pi}{2} \cdot \frac{1 \cdot 3 \cdots (2k-1)}{2 \cdot 4 \cdots (2k)} \quad \int_0^{\frac{\pi}{2}} \sin^{2k+1} x \, dx = \frac{2 \cdot 4 \cdots (2k)}{3 \cdot 5 \cdots (2k+1)}$$

$$\iint f(x, y) \, dx \, dy = \iint f(g(u, v), h(u, v)) \left| \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u} \right| \, du \, dv$$

$$\text{polar: } \iint r \cdot f(r \cos \theta, r \sin \theta) \, dr \, d\theta$$

$$\text{spherical: } \iiint \rho^2 \sin \phi \cdot f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \theta) \, d\phi \, d\theta \, d\rho$$

$$\text{Green's theorem: } \iint \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \, dx \, dy = \oint (P \, dx + Q \, dy)$$

Area: $A = \oint x \, dy = -\oint y \, dx$ (counter-clockwise)

Circular sector: $(x, y) = (x_0 + r \cos \theta, y_0 + r \sin \theta)$,

$$\int_{\alpha}^{\beta} x \, dy - y \, dx = r(r\theta + x_0 \sin \theta - y_0 \cos \theta) \Big|_{\alpha}^{\beta}$$

6.17 Floor Sum

$$m = \lfloor \frac{a+b}{c} \rfloor$$

$$\text{Time complexity: } O(\log n)$$

$$f(a, b, c, n) = \sum_{i=0}^n \lfloor \frac{ai+b}{c} \rfloor$$

$$= \begin{cases} \lfloor \frac{a}{c} \rfloor \cdot \frac{n(n+1)}{2} + \lfloor \frac{b}{c} \rfloor \cdot (n+1) \\ + f(a \bmod c, b \bmod c, c, n), & a \geq c \vee b \geq c \\ 0, & n < 0 \vee a = 0 \\ nm - f(c, c-b-1, a, m-1), & \text{otherwise} \end{cases}$$

$$g(a, b, c, n) = \sum_{i=0}^n i \lfloor \frac{ai+b}{c} \rfloor$$

$$= \begin{cases} \lfloor \frac{a}{c} \rfloor \cdot \frac{n(n+1)(2n+1)}{6} + \lfloor \frac{b}{c} \rfloor \cdot \frac{n(n+1)}{2} \\ + g(a \bmod c, b \bmod c, c, n), & a \geq c \vee b \geq c \\ 0, & n < 0 \vee a = 0 \\ \frac{1}{2} \cdot (n(n+1)m - f(c, c-b-1, a, m-1) \\ - h(c, c-b-1, a, m-1)), & \text{otherwise} \end{cases}$$

$$h(a, b, c, n) = \sum_{i=0}^n \left\lfloor \frac{ai+b}{c} \right\rfloor^2$$

$$= \begin{cases} \left\lfloor \frac{a}{c} \right\rfloor^2 \cdot \frac{n(n+1)(2n+1)}{6} + \left\lfloor \frac{b}{c} \right\rfloor^2 \cdot (n+1) \\ + \left\lfloor \frac{a}{c} \right\rfloor \cdot \left\lfloor \frac{b}{c} \right\rfloor \cdot n(n+1) \\ + h(a \bmod c, b \bmod c, c, n) \\ + 2 \left\lfloor \frac{a}{c} \right\rfloor \cdot g(a \bmod c, b \bmod c, c, n) \\ + 2 \left\lfloor \frac{b}{c} \right\rfloor \cdot f(a \bmod c, b \bmod c, c, n), & a \geq c \vee b \geq c \\ 0, & n < 0 \vee a = 0 \\ nm(m+1) - 2g(c, c-b-1, a, m-1) \\ - 2f(c, c-b-1, a, m-1) - f(a, b, c, n), & \text{otherwise} \end{cases}$$

7 Polynomial

7.1 FFT [43cc28]

```
const double pi=acos(-1);
typedef complex<double> cp;
const int N=(1<<17);
struct FFT
{
    // n has to be same as a.size()
    int n, rev[N];
    cp omega[N], iomega[N];
    void init(int _n) {
        n=_n;
        for(int i=0; i<n; i++) {
            omega[i]=cp(cos(2*pi/n*i), sin(2*pi/n*i));
            iomega[i]=conj(omega[i]);
        }
        int k=log2(n);
        for(int i=0; i<n; i++) {
            rev[i]=0;
            for(int j=0; j<k; j++) if(i&(1<<j))
                rev[i]|=(1<<(k-j-1));
        }
    }
    void tran(vector<cp> &a, cp* xomega)
    {
        for(int i=0; i<n; i++) if(i<rev[i])
            swap(a[i], a[rev[i]]);
        for(int len=2; len<=n; len<=1) {
            int mid=len>>1, r=n/len;
            for(int j=0; j<n; j+=len) {
                for(int i=0; i<mid; i++) {
                    cp t=xomega[r*i]*a[j+mid+i];
                    a[j+mid+i]=a[j+i]-t;
                    a[j+i]+=t;
                }
            }
        }
    }
    void fft(vector<cp> &a) {tran(a, omega);}
    void ifft(vector<cp> &a) {
        tran(a, iomega);
        for(int i=0; i<n; i++) a[i]/=n;
    }
};
```

7.2 NTT [39f8b1]

```
//(2^16)+1, 65537, 3
//7*17*(2^23)+1, 998244353, 3
//1255*(2^20)+1, 1315962881, 3
//51*(2^25)+1, 1711276033, 29
// only works when sz(A) + sz(B) - 1 <= MAXN
// implement add, po, mul, inv first
template<int N, int RT>
struct NTT{
    int w[N];
    NTT() {
        int dw = po(RT, (mod-1) / N);
        w[0] = 1;
        for(int i=1; i<N; i++) w[i] = mul(w[i-1], dw);
    }
    void bitrev(vector<int>& a, int n){
        int i=0;
        for(int j=1; j<n-1; j++) {
            for(int k=n>>1; (i^=k) < k; k>>=1);
            if(j<i) swap(a[i], a[j]);
        }
    }
```

```
}
void operator()(vector<int>& a, int n, bool ix = false)
{
    bitrev(a, n);
    for(int L=2; L<=n; L<=1){
        int dx = N/L, dl = L>>1;
        for(int i=0; i<n; i+=L){
            for(int j=i, x=0; j<i+dl; j++, x+=dx){
                int tmp = mul(a[j+dl], w[x]);
                a[j+dl] = a[j];
                add(a[j+dl], -tmp);
                add(a[j], tmp);
            }
        }
    }
    if(ix){
        reverse(a.begin()+1, a.begin()+n);
        int invn = inv(n);
        for(int i=0; i<n; i++) a[i] = mul(a[i], invn);
    }
};
```

7.3 FWT [d4ce4e]

```
/* (op, invop): (oyx: op from y to x)
or: (1, 0, 1, 1), (1, 0, -1, 1)
and: (1, 1, 0, 1), (1, -1, 0, 1)
xor: (1, 1, 1, -1), (1/2, 1/2, 1/2, -1/2) */
void fwt(vector<int>& a, int n, int oxx, int oyx, int
    oxy, int oyy){ // 2^20, 0.5s
    for(int L=2; L<=n; L<=1)
        for(int i=0; i<n; i+=L)
            for(int j=i; j<i+(L>>1); j++){
#define x a[j]
#define y a[j+(L>>1)]
                tie(x, y) = pii(mul(x, oxx)+mul(y, oyx), mul(x,
                    oxy)+mul(y, oyy));
                add(x, 0); add(y, 0);
#undef x
#undef y
            }
    }
    vector<int> subset_conv(vector<int> a, vector<int> b,
        int L){
        int n = 1<<L; // 2^20, 5.5s, c-style arrays -> 2.1s
        vector<vector<int>> f(L+1, vector<int>(n)), g=f, h=f;
        vector<int> ct(n), c(n);
        for(int i=1; i<n; i++) ct[i] = ct[i&(i-1)] + 1;
        for(int i=0; i<n; i++) { f[ct[i]][i] = a[i]; g[ct[i]][i]
            = b[i]; }
        for(int i=0; i<=L; i++) { fwt(f[i], n, 1, 0, 1, 1); fwt
            (g[i], n, 1, 0, 1, 1); }
        for(int i=0; i<=L; i++)
            for(int j=0; j<=i; j++)
                for(int x=0; x<n; x++)
                    add(h[i][x], mul(f[j][x], g[i-j][x]));
        for(int i=0; i<=L; i++) fwt(h[i], n, 1, 0, -1, 1);
        for(int i=0; i<n; i++) c[i] = h[ct[i]][i];
        return c;
    }
```

7.4 Polynomial [ede4d0]

```
// may need fac ivf
// == ab8066 ==
#define fi(s, n) for (int i = (int)(s); i < (int)(n); i++)
#define neg(x) (x ? mod - x : 0)
#define V (*this)
template<int MAXN, int RT> // MAXN = 2^k
struct Poly : vector<int> { // coefficients in [0, P)
    using vector<int>::vector;
    static inline NTT<MAXN, RT> ntt;
    int n() const { return (int)size(); } // n() >= 1
    Poly(const Poly &p, int m) : vector<int>(m) { copy_n(
        p.data(), min(p.n(), m), data()); }
    Poly &irev() { return reverse(data(), data() + n()),
        V; }
    Poly &isz(int m) { return resize(m), V; }
    // == e33faa ==
    Poly &iadd(const Poly &rhs) { // 1c6277
```

```

    fi(0, n()) add(V[i], rhs[i]);
    return V; // need n() == rhs.n()
}
Poly &imul(int k) { // 7e5b36
    fi(0, n()) V[i] = mul(V[i], k);
    return V;
}
Poly &mul_xk(int m) { // 19862a
    if(m<0) erase(begin(), begin()+(-m));
    else if(m>0) insert(begin(), m, 0);
    return V;
}
Poly Mul(const Poly &rhs) const { // ecd03e
    int m = 1;
    while (m < n() + rhs.n() - 1) m <= 1;
    assert(m <= MAXN);
    Poly X(V, m), Y(rhs, m);
    ntt(X, m); ntt(Y, m);
    fi(0, m) X[i] = mul(X[i], Y[i]);
    ntt(X, m, true);
    return X.isz(n() + rhs.n() - 1);
}
Poly Inv() const { // 77f977
    if (n() == 1) return {inv(V[0])};
    int m = 1; // need V[0] != 0, 2*sz<=MAXN
    while (m < n() * 2) m <= 1;
    assert(m <= MAXN);
    Poly Xi = Poly(V, (n() + 1) / 2).Inv().isz(m);
    Poly Y(V, m);
    ntt(Xi, m); ntt(Y, m);
    fi(0, m) {
        Xi[i] = mul(Xi[i], (2 - mul(Xi[i], Y[i])));
        add(Xi[i], mod);
    }
    ntt(Xi, m, true);
    return Xi.isz(n());
}
// == 095701 ==
Poly &shift_inplace(const int &c) { // need fac[],
    ivf[]
    int n = V.n(); // 2 * sz <= MAXN
    for (int i = 0; i < n; i++) V[i] = mul(V[i], fac[i]);
    Poly g(n);
    int cp = 1;
    for (int i = 0; i < n; i++){
        g[i] = mul(cp, ivf[i]);
        cp = mul(cp, c);
    }
    V = V.irev().Mul(g).isz(n).irev();
    for (int i = 0; i < n; i++) V[i] = mul(V[i], ivf[i]);
    return V;
}
Poly Shift(const int &c) const { return Poly(V).
    shift_inplace(c); }
// == 0bd61d ==
Poly Dx() const {
    Poly ret(n() - 1);
    fi(0, ret.n()) ret[i] = mul(i+1, V[i+1]);
    return ret.isz(max(1, ret.n()));
}
Poly Sx() const {
    Poly ret(n() + 1);
    fi(0, n()) ret[i + 1] = mul(inv(i+1), V[i]);
    return ret;
}
// == 10e23d ==
Poly Ln() const { // V[0] == 1, 2*sz<=MAXN
    return Dx().Mul(Inv()).Sx().isz(n());
}
Poly Exp() const { // V[0] == 0, 2*sz<=MAXN
    if (n() == 1) return {1};
    Poly X = Poly(V, (n() + 1) / 2).Exp().isz(n());
    Poly Y = X.Ln();
    Y[0] = mod - 1;
    fi(0, n()){
        Y[i] = V[i] - Y[i];
        add(Y[i], mod);
    }
    return X.Mul(Y).isz(n());
}
// == eaf14c ==
// M := P(P - 1). If k >= M, k := k % M + M.
Poly Pow(int k) const { // 2*sz<=MAXN
    int nz = 0;
    while (nz < n() && !V[nz]) nz++;
    if (nz * min(k, (11)n()) >= n()) return Poly(n());
    if (!k) return Poly(Poly{1}, n());
    Poly X(data() + nz, data() + nz + n() - nz * k);
    const int c = po(X[0], k % (mod - 1));
    return X.Ln().imul(k % mod).Exp().imul(c).irev().
        isz(n()).irev();
}
// == cdf741 ==
Poly _tmul(int nn, const Poly &rhs) const {
    Poly Y = Mul(rhs).isz(n() + nn - 1);
    return Poly(Y.data() + n() - 1, Y.data() + Y.n());
}
vector<int> _eval(const vector<int> &x, const vector<
    Poly> &up) const { // 82d6be
    const int m = (int)x.size();
    if (!m) return {};
    vector<Poly> down(m * 2);
    down[1] = Poly(up[1]).irev().isz(n()).Inv().irev().
        _tmul(m, V);
    fi(2, m * 2) down[i] = up[i ^ 1]._tmul(up[i].n() -
        1, down[i / 2]);
    vector<int> y(m);
    fi(0, m) y[i] = down[m + i][0];
    return y;
}
static vector<Poly> _tree1(const vector<int> &x) { //
    2a0b6b
    const int m = (int)x.size();
    vector<Poly> up(m * 2);
    fi(0, m) up[m + i] = {neg(x[i]), 1};
    for (int i = m - 1; i > 0; --i) up[i] = up[i * 2].
        Mul(up[i * 2 + 1]);
    return up;
}
vector<int> Eval(const vector<int> &x) const {
    auto up = _tree1(x); // 2^17, 1.8s
    return _eval(x, up);
}
// == f9ecd4 ==
static Poly Interpolate(const vector<int> &x, const
    vector<int> &y) { // 8f2a08
    const int m = (int)x.size(); // 2^17, 2.3s
    vector<Poly> up = _tree1(x), down(m * 2);
    vector<int> z = up[1].Dx()._eval(x, up);
    fi(0, m) z[i] = mul(y[i], inv(z[i]));
    fi(0, m) down[m + i] = {z[i]};
    for (int i = m - 1; i > 0; --i)
        down[i] = down[i * 2].Mul(up[i * 2 + 1]).iadd(
            down[i * 2 + 1].Mul(up[i * 2]));
    return down[1];
}
pair<Poly, Poly> DivMod(const Poly &rhs) const { //
    a75170
    if (n() < rhs.n()) return {{0}, V}; // 5e5, 0.9s
    const int m = n() - rhs.n() + 1;
    Poly X(rhs); // (rhs).back() != 0
    X.irev().isz(m);
    Poly Y(V); Y.irev().isz(m);
    Poly Q = Y.Mul(X.Inv()).isz(m).irev();
    X = rhs.Mul(Q), Y = V;
    fi(0, n()) add(Y[i], -X[i]);
    return {Q, Y.isz(max(1, rhs.n() - 1))};
}
// == 7cd4c4 ==
Poly _Sqrt() const { // Jacobi(V[0], P) = 1
    if (n() == 1) return {QuadraticResidue(V[0], mod)};
    Poly X = Poly(V, (n() + 1) / 2)._Sqrt().isz(n());
    return X.iadd(Mul(X.Inv()).isz(n())).imul(mod / 2 +
        1);
}
Poly Sqrt() const { // 288427
    Poly a; // 2 * sz <= MAXN
    bool has = 0;
    for (int i = 0; i < n(); i++) {
        if (V[i]) has = 1;
        if (has) a.push_back(V[i]);
    }
}

```

```

    if (!has) return V;
    if ((n() + a.n()) % 2 || Jacobi(a[0], mod) != 1) {
        return Poly();
    }
    a = a.isz((n() + a.n()) / 2)._Sqrt();
    int sz = a.n();
    a.isz(n());
    rotate(a.begin(), a.begin() + sz, a.end());
    return a;
}
// == 0a54cf ==
Poly Shift_samples(int c, int m){
    // V = \sum f(n) x^n, return f(c), ..., f(c+m-1);
    // 2^19, 2s
    Poly A=V;
    Poly Q(n()+1), S(n());
    int nw=1;
    fi(0, n()+1) { Q[i] = mul(1-2*(i&1), nw); nw = mul(
        nw, n()-i, inv(i+1)); }
    nw=1;
    fi(0, n()) { S[i] = mul(1-2*(i&1), nw); nw = mul(nw,
        c-i, inv(i+1)); }
    S=S.Shift(1);
    fi(0, n()) if(i&1) { S[i] = mul(S[i], -1); add(S[i],
        mod); }
    Poly C=Mul(Q).mul_xk(-n());
    auto tmp=Q.isz(m).Inv();
    C=C.imul(-1).Mul(tmp).isz(m).mul_xk(n());
    A=A.isz(n()+m).iadd(C).irev().isz(n()+m);
    return A.Mul(S).mul_xk(-n()+1).isz(m+1).irev();
}
// == 224b20 ==
Poly power_projection(Poly wt, int m) { // 857237
    assert(n() == wt.n()); // 4*sz <= MAXN
    if (!n()) return Poly(m + 1, 0);
    if (V[0] != 0) {
        int c = V[0];
        V[0] = 0;
        Poly A = V.power_projection(wt, m);
        fi(0, m + 1) A[i] = mul(A[i], fac[i]);
        Poly B(m + 1);
        int pow = 1;
        fi(0, m + 1) { B[i] = mul(pow, ivf[i]); pow = mul(
            pow, c); } // inv. of fac
        A = A.Mul(B).isz(m + 1);
        fi(0, m + 1) A[i] = mul(A[i], fac[i]);
        return A;
    }
    int n = 1;
    while (n < V.n()) n *= 2;
    isz(n), wt.isz(n).irev();
    int k = 1;
    Poly p(wt, 2 * n), q(V, 2 * n);
    q.imul(mod - 1);
    while (n > 1) {
        Poly r(2 * n * k);
        fi(0, 2 * n * k) r[i] = (i % 2 == 0 ? q[i] : neg(
            q[i]));
        Poly pq = p.Mul(r).isz(4 * n * k);
        Poly qq = q.Mul(r).isz(4 * n * k);
        fi(0, 2 * n * k) {
            add(pq[2 * n * k + i], p[i]);
            add(qq[2 * n * k + i], (q[i] + r[i]) % mod);
        }
        fill(p.begin(), p.end(), 0);
        fill(q.begin(), q.end(), 0);
        for(int j = 0; j < 2 * k; j++) fi(0, n / 2) {
            p[n * j + i] = pq[(2 * n) * j + (2 * i + 1)];
            q[n * j + i] = qq[(2 * n) * j + (2 * i + 0)];
        }
        n /= 2; k *= 2;
    }
    Poly ans(k);
    fi(0, k) ans[i] = p[2 * i];
    return ans.irev().isz(m + 1);
}
Poly FPSinv() { // 49e5ad
    const int n = V.n() - 1; // 2^17, 4s
    if (n == -1) return {};
    assert(V[0] == 0);
    if (n == 0) return V;
    assert(V[1] != 0);

```

```

    int c = V[1], ic = inv(c);
    imul(ic);
    Poly wt(n + 1);
    wt[n] = 1;
    Poly A = V.power_projection(wt, n);
    Poly g(n);
    fi(1, n + 1) g[n - i] = mul(n, A[i], inv(i));
    g = g.Pow(neg(inv(n)));
    g.insert(g.begin(), 0);
    int pow = 1;
    fi(0, g.n()) { g[i] = mul(g[i], pow); pow = mul(pow,
        ic); }
    return g;
}
// == 71ee33 ==
Poly TMul(const Poly &rhs) const { // this[i] - rhs[j]
    j = k;
    return Poly(V.irev().Mul(rhs).isz(n()).irev());
}
Poly comp_rec(int n, int k, Poly Q) { // 06ca07
    if (n == 1) {
        Poly p(2 * k);
        irev();
        fi(0, k) p[2 * i] = V[i];
        return p;
    }
    Poly R(2 * n * k);
    fi(0, 2 * n * k) R[i] = (i % 2 == 0 ? Q[i] : neg(Q[
        i]));
    Poly QQ = Q.Mul(R).isz(4 * n * k);
    fi(0, 2 * n * k)
        add(QQ[2 * n * k + i], (Q[i] + R[i]) % mod);
    Poly nxt_Q(2 * n * k);
    for(int j = 0; j < 2 * k; j++) fi(0, n / 2)
        nxt_Q[n * j + i] = QQ[(2 * n) * j + (2 * i + 0)];
    Poly nxt_p = comp_rec(n / 2, k * 2, nxt_Q);
    Poly pq(4 * n * k);
    for(int j = 0; j < 2 * k; j++) fi(0, n / 2)
        add(pq[(2 * n) * j + (2 * i + 1)], nxt_p[n * j +
            i]);
    Poly p(2 * n * k);
    fi(0, 2 * n * k) add(p[i], pq[2 * n * k + i]);
    pq.pop_back();
    Poly x = pq.TMul(R);
    fi(0, 2 * n * k) add(p[i], x[i]);
    return p;
}
Poly FPScomp(Poly g) { // solves V(g(x))
    int sz = 1; // 2^17, 5s
    while(sz < n() || sz < g.n()) sz <= 1;
    return isz(sz), comp_rec(sz, 1, g.imul(mod-1).isz(2
        * sz)).isz(sz).irev();
}
// == e42b36 ==
};
#undef V
#undef neg
#undef fi
using Poly_t = Poly<1 << 21, 3>;

```

7.5 BM-BM [362501]

```

// out[i] = me[0]*out[i-1] + me[1]*out[i-2]+...
vector<int> BerlekampMassey(const vector<int> &output)
{ // 2e7709
    vector<int> d(output.size() + 1), me, he;
    for (int f = 0, i = 1; i <= sz(output); ++i) {
        for (int j = 0; j < sz(me); ++j)
            add(d[i], mul(output[i - j - 2], me[j]));
        add(d[i], -output[i-1]);
        if (d[i] == 0) continue;
        if (me.empty()) {
            me.resize(f = i);
            continue;
        }
        vector<int> o(i - f - 1);
        int k = mul(d[i], -inv(d[f]));
        o.push_back(-k);
        for (auto x : he) o.push_back(mul(x, k));
        if (o.size() < me.size()) o.resize(me.size());
        for (size_t j = 0; j < me.size(); ++j) add(o[j], me
            [j]);
    }
}

```

```

    if (i-f+he.size() >= me.size()) he = me, f = i;
    me = o;
}
for(auto &h: me) add(h, mod);
return me;
}

// Finds the k-th coefficient of P / Q in O(d log d log k)
// size of NTT has to > 2 * d
NTT<1<<19, 3> ntt;
int BostanMori(vector<int> P, vector<int> Q, long long k) { // 1bee33
    int d = max((int)P.size(), (int)Q.size() - 1);
    vector M = {P, Q};
    M[0].resize(d, 0);
    M[1].resize(d + 1, 0);
    int sz = (2 * d + 1 == 1 ? 2 : (1 << (1 + log2(d * 2) + 1)));
    vector<int> Qn(sz);
    vector N(2, vector<int>(sz));
    while(k) {
        fill(iter(Qn), 0);
        for(int i = 0; i < d + 1; i++){
            Qn[i] = M[1][i] * ((i & 1) ? -1 : 1);
            add(Qn[i], mod);
        }
        ntt(Qn, sz, false);
        int t[2] = {(int)k & 1, 0};
        for(int i = 0; i < 2; i++){
            fill(iter(N[i]), 0);
            copy(iter(M[i]), N[i].begin());
            ntt(N[i], sz, false);
            for(int j = 0; j < sz; j++) N[i][j] = mul(N[i][j], Qn[j]);
            ntt(N[i], sz, true);
            for(int j = t[i]; j < 2 * sz(M[i]); j += 2){
                M[i][j >> 1] = N[i][j];
            }
        }
        k >>= 1;
    }
    return mul(M[0][0], inv(M[1][0]));
}

// a_n = \sum_{j=1}^d a_{n-j} c_j, c_0 = 0
int LinearRecursion(vector<int> a, vector<int> c, ll k)
{ // 3b1227
    int d = sz(a), sz = 1; // 1e5, 5s
    while(sz <= 2*d) sz <<= 1;
    c[0] = mod - 1;
    for(auto &i : c) i = i ? mod - i : 0;
    auto A = a; A.resize(sz);
    auto C = c; C.resize(sz);
    ntt(A, sz, false), ntt(C, sz, false);
    for(int i = 0; i < sz; i++) A[i] = mul(A[i], C[i]);
    ntt(A, sz, true);
    A.resize(d);
    return BostanMori(A, c, k);
}

```

7.6 Generating Functions

- Ordinary Generating Function $A(x) = \sum_{i \geq 0} a_i x^i$

$$\begin{aligned}
 & A(rx) \Rightarrow r^n a_n \\
 & A(x) + B(x) \Rightarrow a_n + b_n \\
 & A(x)B(x) \Rightarrow \sum_{i=0}^n a_i b_{n-i} \\
 & A(x)^k \Rightarrow \sum_{i_1+i_2+\dots+i_k=n} a_{i_1} a_{i_2} \dots a_{i_k} \\
 & xA(x)' \Rightarrow na_n \\
 & \frac{A(x)}{1-x} \Rightarrow \sum_{i=0}^n a_i
 \end{aligned}$$

- Exponential Generating Function $A(x) = \sum_{i \geq 0} \frac{a_i}{i!} x^i$

$$\begin{aligned}
 & A(x) + B(x) \Rightarrow a_n + b_n \\
 & A^{(k)}(x) \Rightarrow a_{n+k} \\
 & A(x)B(x) \Rightarrow \sum_{i=0}^n \binom{n}{i} a_i b_{n-i} \\
 & A(x)^k \Rightarrow \sum_{i_1+i_2+\dots+i_k=n} \binom{n}{i_1, i_2, \dots, i_k} a_{i_1} a_{i_2} \dots a_{i_k} \\
 & xA(x) \Rightarrow na_n
 \end{aligned}$$

- Special Generating Function

$$\begin{aligned}
 & (1+x)^n = \sum_{i \geq 0} \binom{n}{i} x^i \\
 & \frac{1}{(1-x)^n} = \sum_{i \geq 0} \binom{n+i-1}{n-1} x^i
 \end{aligned}$$

- Catalan number: $f(x) = \frac{1-\sqrt{1-4x}}{2x}$

- Suppose $A(x)$ is the EGF of a labelled component. Then $B(x) = e^{A(x)}$ is the EGF of the same thing but with any number of components.

- Lagrange's Inversion Formula

$F(G(x)) = G(F(x)) = x$, then $F(0) = G(0) = 0$ and $F'(0), G'(0) \neq 0$. Let H be any FPS, then

$$n[x^n]F(x) = [x^{n-1}]H'(x) \left(\frac{x}{G(x)} \right)^n.$$

- Newton's Method

Suppose FPS P satisfies $F(x, P(x)) = 0$ where $F(x, y)$ is some polynomial, then

$$P_{2n}(x) = P_n(x) - F(x, P_n(x)) / \frac{\partial F}{\partial y}(x, P_n(x)).$$

8 Geometry

8.1 Basic [d4b33e]

```

// == d2c7dd ==
struct PT {
    ll x, y;
    auto operator<=>(const PT&) const = default;
};
PT operator+(PT p1, PT p2)
{ return PT{p1.x + p2.x, p1.y + p2.y}; }
PT operator-(PT p1, PT p2)
{ return PT{p1.x - p2.x, p1.y - p2.y}; }
ll operator*(PT p1, PT p2)
{ return p1.x * p2.x + p1.y * p2.y; }
ll operator^(PT p1, PT p2)
{ return p1.x * p2.y - p1.y * p2.x; }
int sign(ll a) { return a == 0 ? 0 : a > 0 ? 1 : -1; }
int ori(PT a, PT b, PT c) { // is C to the left of A->B
    return sign((b-a)^(c-a)); }
bool btw(PT a, PT b, PT c) { // is C between AB
    return !ori(a, b, c) && sign((a-c)*(b-c)) <= 0; }
PT ccw90(PT p) { return PT(-p.y, p.x); }
// == f61f45 ==
// change PT::x, y to FP type, replace sign()
// needs PT+-PT, PT*/scalar
const double EPS = 1e-9;
int sign(double a) { return abs(a)<EPS?0:a>0?1:-1; }
double abs2(PT p){ return p * p; }
double abs(PT p){ return sqrt(p * p); }
PT proj(PT a, PT b, PT c){ // C projected to AB
    return a+(b-a)*((c-a)*(b-a)/abs2(b-a)); }
double dist(PT a,PT b,PT c){ // distance from C to AB
    return abs((c-a)^(b-a))/abs(b-a); }
// struct Line{ PT a, b; };
// struct Cir{ PT o; double r; };

```

8.2 Convex Hull [903417]

```

// (int) requires sz(pt) >= 2 & distinct points
// points on edges are excluded
vector<PT> ConvexHull(vector<PT> pt) {
    int n = sz(pt);
    sort(iter(pt));
    vector<PT> ans{pt[0]};
    For(t, 0, 1) {
        int m = sz(ans);
        For(i, 1, n - 1) {
            while (sz(ans) > m && ori(ans[sz(ans) - 2],
                ans.back(), pt[i]) <= 0) ans.pop_back();
            ans.push_back(pt[i]);
        }
        reverse(iter(pt));
    }
    if (sz(ans) > 1) ans.pop_back();
    return ans;
}

```

8.3 Dynamic Convex Hull [95890b]

```
// (int)
struct DynamicConvexHull {
    struct UpCmp {
        bool operator()(PT a, PT b) const {
            return a < b;
        }
    };
    struct DownCmp {
        bool operator()(PT a, PT b) const {
            return a > b;
        }
    };
    template <typename T>
    struct Hull {
        set<PT,T> hull;
        bool chk(PT i, PT j, PT k) {
            return ((k-i)^(j-i)) > 0;
        }
        void insert(PT x) {
            if(inside(x)) return;
            hull.insert(x);
            auto it=hull.lower_bound(x);
            if(next(it)!=hull.end()) {
                for (auto it2 = next(it);
                    next(it2) != hull.end(); ++it2) {
                    if(chk(x,*it2,*next(it2))) break;
                    hull.erase(it2);
                    it2=hull.lower_bound(x);
                }
            }
            it=hull.lower_bound(x);
            if(it!=hull.begin()) {
                for (auto it2 = prev(it);
                    it2 != hull.begin(); --it2) {
                    if(chk(*prev(it2),*it2,x)) break;
                    hull.erase(it2);
                    it2=hull.lower_bound(x);
                    if(it2==hull.begin()) break;
                }
            }
        }
        bool inside(PT x) {
            if (hull.lower_bound(x) != hull.end() &&
                *hull.lower_bound(x) == x)
                return true;
            auto it=hull.lower_bound(x);
            bool ans=false;
            if(it!=hull.begin()&&it!=hull.end()) {
                ans=!chk(*prev(it),x,*it);
            }
            return ans;
        }
    };
    Hull<UpCmp> up;
    Hull<DownCmp> down;
    void insert(PT x){up.insert(x),down.insert(x);}
    bool inside(PT x) {
        return up.inside(x) && down.inside(x);
    }
};
```

8.4 Point In Convex Hull [9a1f2c]

```
// (int) C must be a convex polygon in cyclic order
bool PointInConvex(const vector<PT> &C, PT p,
    bool strict = true) {
    int a = 1, b = int(C.size()) - 1, r = !strict;
    if (C.size() == 0) return false;
    if (C.size() < 3) return r && btw(C[0], C.back(), p);
    if (ori(C[0], C[a], C[b]) > 0) swap(a, b);
    if (ori(C[0], C[a], p) >= r ||
        ori(C[0], C[b], p) <= -r) return false;
    while (abs(a - b) > 1) {
        int c = (a + b) / 2;
        (ori(C[0], C[c], p) > 0 ? b : a) = c;
    }
    return ori(C[a], C[b], p) < r;
}
```

8.5 Point In Circle [9d9594]

```
// (int) p1, p2, p3 must be in CCW order
```

```
// test if p4 strictly in circumscribed circle of p1p2p3
bool InCir(PT p1, PT p2, PT p3, PT p4) {
    p1 = p1 - p4; p2 = p2 - p4; p3 = p3 - p4;
    ll l1 = p1 * p1, l2 = p2 * p2, l3 = p3 * p3;
    i128 det = (i128)l1 * (p2 ^ p3)
        + (i128)l2 * (p3 ^ p1)
        + (i128)l3 * (p1 ^ p2);
    return det > 0;
}
```

8.6 Half Plane Intersection [c77d62]

```
// (int) needs polar angle
// arr must be nonempty; keeps the left of each line
auto area_pair(Line a, Line b) {
    return make_pair((a.b - a.a) ^ (b.a - a.a),
        (a.b - a.a) ^ (b.b - a.a));
}
bool isin(Line l0, Line l1, Line l2) {
    // Check inter(l1, l2) strictly in l0
    auto [a02X, a02Y] = area_pair(l0, l2);
    auto [a12X, a12Y] = area_pair(l1, l2);
    if (a12X < a12Y) a12X *= -1, a12Y *= -1;
    return (i128)a02Y * a12X > (i128)a02X * a12Y;
}
/* [solution exists] <=> [result.size() > 2] */
/* --^-- Line.a --^-- Line.b --^-- */
vector<Line> HalfPlaneInter(vector<Line> arr) {
    sort(arr.begin(), arr.end(), [&](Line a, Line b) {
        PT p1 = a.b - a.a, p2 = b.b - b.a;
        if (polar(p1, p2) != 0) return polar(p1, p2) < 0;
        return ori(a.a, a.b, b.b) < 0;
    });
    deque<Line> dq(1, arr[0]);
    auto pop_back = [&](int t, Line p) {
        while (sz(dq) >= t &&
            !isin(p, dq[sz(dq) - 2], dq.back()))
            dq.pop_back();
    };
    auto pop_front = [&](int t, Line p) {
        while (sz(dq) >= t && !isin(p, dq[0], dq[1]))
            dq.pop_front();
    };
    for (auto p : arr)
        if (polar(dq.back().b - dq.back().a,
            p.b - p.a) != 0)
            pop_back(2, p), pop_front(2, p),
            dq.push_back(p);
    pop_back(3, dq[0]), pop_front(3, dq.back());
    return vector<Line>(dq.begin(), dq.end());
}
```

8.7 Minkowski Sum [f559ee]

```
// (int) needs convex hull
// each input needs at least two distinct points
vector<PT> Minkowski(vector<PT> a, vector<PT> b) {
    a = ConvexHull(a), b = ConvexHull(b);
    int n = sz(a), m = sz(b);
    vector<PT> c{a[0] + b[0]}, s1, s2;
    For(i, 0, n - 1)
        s1.push_back(a[(i + 1) % n] - a[i]);
    For(i, 0, m - 1)
        s2.push_back(b[(i + 1) % m] - b[i]);
    for (int p1 = 0, p2 = 0; p1 < n || p2 < m;)
        if (p2 == m || (p1 < n &&
            sign(s1[p1] ^ s2[p2]) >= 0))
            c.push_back(c.back() + s1[p1++]);
        else c.push_back(c.back() + s2[p2++]);
    return ConvexHull(c);
}
```

8.8 Polar Angle [423cf8]

```
// (int) CCW from (1, 0), without length tie-breaking
int halfplane(PT p) {
    if (sign(p * p) == 0) return 0;
    bool upper = sign(p.y) > 0 ||
        (sign(p.y) == 0 && sign(p.x) > 0);
    return 1 - 2 * upper;
}
// upper(-1) -> origin(0) -> Lower(1)
strong_ordering polar(PT a, PT b) {
```

```

int ha = halfplane(a), hb = halfplane(b);
if (ha != hb) return ha <= hb;
return 0 <= sign(a ^ b);
}

```

8.9 Rotating Sweep Line [f1c4b0]

```

// (int) needs polar angle
// pts: 0-indexed Pt array
void RotSweepLine(int n, PT* pts) {
    using E = pair<PT, pii>;
    vector<E> ev; // dir, i, j: (i, j) => (j, i)
    For(i, 0, n - 1) For(j, 0, i - 1) {
        PT dir = pts[j] - pts[i];
        halfplane(dir) < 0 ? ev.push_back({dir, {i, j}})
        : ev.push_back({PT(0, 0) - dir, {j, i}});
    }
    sort(ev.begin(), ev.end(), [&](E e1, E e2) {
        auto pol = polar(e1.F, e2.F);
        return pol < 0 || (pol == 0 &&
            pll(e1.F * pts[e1.S.F], e1.F * pts[e1.S.S])
            < pll(e1.F * pts[e2.S.F], e1.F * pts[e2.S.S]));
    });
    vector<int> ord(n), rk(n);
    iota(ord.begin(), ord.end(), 0);
    sort(ord.begin(), ord.end(), [&](int i, int j) {
        return make_pair(pts[i].y, pts[i].x)
            < make_pair(pts[j].y, pts[j].x);
    });
    For(i, 0, n - 1) rk[ord[i]] = i;
    // init order with point indices ord[*]
    int ne = (int)ev.size();
    For(ie, 0, ne - 1) {
        int i, j; tie(i, j) = ev[ie].S;
        // update swap i, j
        rk[i]++; rk[j]--; // rk[point idx]: rank in ord
        tie(ord[rk[i]], ord[rk[i] - 1]) = tie(i, j);
        if (ie == ne - 1 ||
            polar(ev[ie + 1].F, ev[ie].F) != 0) {
            // update answer
        }
    }
}

```

8.10 Segment Intersect [118283]

```

bool HasInter(PT p1, PT p2, PT p3, PT p4) { // 4d829f
    int a123 = ori(p1, p2, p3), a124 = ori(p1, p2, p4);
    int a341 = ori(p3, p4, p1), a342 = ori(p3, p4, p2);
    if (!a123 && !a124)
        return btw(p1, p2, p3) || btw(p1, p2, p4) ||
            btw(p3, p4, p1) || btw(p3, p4, p2);
    return a123 * a124 <= 0 && a341 * a342 <= 0;
} // (int) does p1p2 intersect p3p4
PT GetInter(PT p1, PT p2, PT p3, PT p4) { // 2629b1
    double a123 = (p2 - p1) ^ (p3 - p1);
    double a124 = (p2 - p1) ^ (p4 - p1);
    return (p4 * a123 - p3 * a124) / (a123 - a124);
} // (FP) C^3 / C^2; requires nonparallel inputs

```

8.11 Circle Intersect With Any [c31e48]

```

// (FP) nondegenerate Lines; polygon edges != 0
vector<PT> CircleLineInter(Cir c, Line l) { // ab01ed
    PT p = l.a + (l.b - l.a) *
        ((c.o - l.a) * (l.b - l.a)) /
        abs2(l.b - l.a);
    double s = (l.b - l.a) ^ (c.o - l.a);
    double h2 = c.r * c.r - s * s /
        abs2(l.b - l.a);
    if (sign(h2) == -1) return {};
    if (sign(h2) == 0) return {p};
    PT h = (l.b - l.a) / abs(l.b - l.a) * sqrt(h2);
    return {p - h, p + h};
}
vector<PT> CirclesInter(Cir c1, Cir c2) { // e9fe3a
    double d2 = abs2(c1.o - c2.o), d = sqrt(d2);
    if (sign(d2) == 0) return {}; // concentric circles
    if (d < max(c1.r, c2.r) - min(c1.r, c2.r) ||
        d > c1.r + c2.r) return {};
    PT u = (c1.o + c2.o) / 2 + (c1.o - c2.o) *
        ((c2.r * c2.r - c1.r * c1.r) / (2 * d2));

```

```

    double A = sqrt((c1.r + c2.r + d) *
        (c1.r - c2.r + d) * (c1.r + c2.r - d) *
        (-c1.r + c2.r + d));
    PT v = PT{c1.o.y - c2.o.y, -c1.o.x + c2.o.x}
        * A / (2 * d2);
    if (sign(v.x) == 0 && sign(v.y) == 0) return {u};
    return {u + v, u - v};
}
double _area(PT pa, PT pb, double r) { // 1238aa
    if (abs(pa) < abs(pb)) swap(pa, pb);
    if (abs(pb) < EPS) return 0;
    double S, h, theta;
    double a = abs(pb), b = abs(pa), c = abs(pb - pa);
    double cosB = pb * (pb - pa) / a / c, B = acos(cosB);
    double cosC = (pa * pb) / a / b, C = acos(cosC);
    if (a > r) {
        S = (C / 2) * r * r;
        h = a * b * sin(C) / c;
        if (h < r && B < numbers::pi / 2)
            S -= acos(h / r) * r * r -
                h * sqrt(r * r - h * h);
    } else if (b > r) {
        theta = numbers::pi - B - asin(sin(B) / r * a);
        S = .5 * a * r * sin(theta) +
            (C - theta) / 2 * r * r;
    } else S = .5 * sin(C) * a * b;
    return S;
}
double AreaPolyCircle(vector<PT> poly, PT O,
    double r) { // 934b82
    double S = 0; int n = sz(poly);
    for (int i = 0; i < n; ++i)
        S += _area(poly[i] - O, poly[(i + 1) % n] - O,
            r) * ori(O, poly[i], poly[(i + 1) % n]);
    return fabs(S);
}

```

8.12 Tangents [b087af]

```

// (FP) circles have positive radii
PT unit(PT p) { return p / abs(p); }
PT ccw(PT p, double t) {
    return PT(
        p.x * cos(t) - p.y * sin(t),
        p.x * sin(t) + p.y * cos(t)
    );
}
vector<Line> Tangent(Cir c, PT p) { // bb4a23
    vector<Line> z;
    double d = abs(p - c.o);
    if (sign(d - c.r) == 0) {
        z.push_back({p, p + ccw90(p - c.o)});
    } else if (d > c.r) {
        double o = acos(c.r / d);
        PT i = unit(p - c.o);
        PT j = ccw(i, o) * c.r;
        PT k = ccw(i, -o) * c.r;
        z.push_back({c.o + j, p});
        z.push_back({c.o + k, p});
    }
    return z;
}
vector<Line> Tangent(Cir c1, Cir c2, int sign1) { //
    sign1 = 1 for outer, -1 for inter // d97c9a
    vector<Line> ret;
    double d_sq = abs2(c1.o - c2.o);
    if (sign(d_sq) == 0) return ret;
    double d = sqrt(d_sq);
    PT v = (c2.o - c1.o) / d;
    double c = (c1.r - sign1 * c2.r) / d;
    if (c * c > 1) return ret;
    double h = sqrt(max(0.0, 1.0 - c * c));
    for (int sign2 = 1; sign2 >= -1; sign2 -= 2) {
        PT n = PT{v.x * c - sign2 * h * v.y,
            v.y * c + sign2 * h * v.x};
        PT p1 = c1.o + n * c1.r;
        PT p2 = c2.o + n * (c2.r * sign1);
        if (sign(p1.x - p2.x) == 0 &&
            sign(p1.y - p2.y) == 0)
            p2 = p1 + ccw90(c2.o - c1.o);
        ret.push_back({p1, p2});
    }
}

```

```

return ret;
}

```

8.13 Tangent to Convex Hull [f1fdc2]

```

// (int) needs cyclic ternary search
// C is nonempty; p is strictly outside C
pii HullTangent(vector<PT> &C, PT p) {
    auto gao = [&](int s) {
        return cyc_tsearch(sz(C), [&](int x, int y)
            { return ori(p, C[x], C[y]) == s; });
    };
    return pii(gao(1), gao(-1));
} // return (a, b), ori(p, C[a], C[b]) >= 0

```

8.14 Minimum Enclosing Circle [07df39]

```

// (FP) needs segment intersect; p must be nonempty
PT circenter(PT a, PT b, PT c) {
    PT ab = (a + b) / 2, ac = (a + c) / 2;
    return GetInter(
        ab, ab + ccw90(b - a),
        ac, ac + ccw90(c - a)
    );
}
Cir MinEnclosing(vector<PT> &p) {
    shuffle(iter(p), mt19937(clock()));
    double r = 0;
    PT cent = p[0];
    For(i, 0, sz(p) - 1) if (abs2(cent - p[i]) > r) {
        cent = p[i]; r = 0;
        For(j, 0, i - 1) if (abs2(cent - p[j]) > r) {
            cent = (p[i] + p[j]) / 2;
            r = abs2(p[j] - cent);
            For(k, 0, j - 1) if (abs2(cent - p[k]) > r) {
                cent = circenter(p[i], p[j], p[k]);
                r = abs2(p[k] - cent);
            }
        }
    }
    return {cent, sqrt(r)};
}

```

8.15 Union of Stuff [ae9b12]

```

// (FP) needs segment intersect
// circles are pairwise distinct; polygons are C CW
vector<pair<double, double>> CoverSegment(Cir a, Cir b)
{ // 7aelbe
    double d = abs(a.o - b.o);
    vector<pair<double, double>> res;
    if (sign(a.r + b.r - d) == 0);
    else if (d <= abs(a.r - b.r) + EPS) {
        if (a.r < b.r)
            res.emplace_back(0, 2 * numbers::pi);
    } else if (d < abs(a.r + b.r) - EPS) {
        double o = acos((a.r * a.r + d * d - b.r * b.r)
            / (2 * a.r * d));
        double z = atan2((b.o - a.o).y, (b.o - a.o).x);
        if (z < 0) z += 2 * numbers::pi;
        double l = z - o, r = z + o;
        if (l < 0) l += 2 * numbers::pi;
        if (r > 2 * numbers::pi) r -= 2 * numbers::pi;
        if (l > r)
            res.emplace_back(l, 2 * numbers::pi),
            res.emplace_back(0, r);
        else res.emplace_back(l, r);
    }
    return res;
}
double CircleUnionArea(vector<Cir> c) { // e653e9
    int n = sz(c);
    double a = 0, w;
    for (int i = 0; w = 0, i < n; ++i) {
        vector<pair<double, double>> s = {
            {2 * numbers::pi, 9}
        }, z;
        for (int j = 0; j < n; ++j) if (i != j) {
            z = CoverSegment(c[i], c[j]);
            for (auto &e : z) s.push_back(e);
        }
        sort(s.begin(), s.end());
    }
}

```

```

auto F = [&] (double t) {
    return c[i].r * (c[i].r * t +
        c[i].o.x * sin(t) - c[i].o.y * cos(t));
};
for (auto &e : s) {
    if (e.first > w) a += F(e.first) - F(w);
    w = max(w, e.second);
}
return a * 0.5;
}
// Union of Polygons
double PolyUnion(vector<vector<PT>> poly) { // 827711
    int n = sz(poly);
    double ans = 0;
    auto solve = [&](PT a, PT b, int cid) {
        vector<pair<PT, int>> event;
        for (int i = 0; i < n; ++i) {
            int st = 0, m = sz(poly[i]);
            while (st < m && ori(poly[i][st], a, b) != 1)
                st++;
            if (st == m) continue;
            For(j, 0, m - 1) {
                PT c = poly[i][(j + st) % m];
                PT d = poly[i][(j + st + 1) % m];
                if (sign((a - b) ^ (c - d)) != 0) {
                    int ok1 = ori(c, a, b) == 1;
                    int ok2 = ori(d, a, b) == 1;
                    if (ok1 ^ ok2)
                        event.emplace_back(GetInter(a, b, c, d),
                            ok1 ? 1 : -1);
                } else if (ori(c, a, b) == 0 &&
                    sign((a - b) * (c - d)) > 0 && i <= cid) {
                    event.emplace_back(c, -1);
                    event.emplace_back(d, 1);
                }
            }
        }
        sort(iter(event), [&](pair<PT, int> i,
            pair<PT, int> j) {
            return (a - i.first) * (a - b) <
                (a - j.first) * (a - b);
        });
        int now = 0;
        PT lst = a;
        for (auto [x, y] : event) {
            if (btw(a, b, lst) && btw(a, b, x) && !now)
                ans += lst ^ x;
            now += y, lst = x;
        }
        For(i, 0, n - 1) For(j, 0, sz(poly[i]) - 1)
            solve(poly[i][j], poly[i][(j + 1) % sz(poly[i])],
                i);
        return ans / 2;
    }
}

```

8.16 Delaunay Triangulation [1c018f]

```

// (int) needs point in circle, segment intersect
const int N = 0; // set per problem constraints
// needs at least one distinct point
/* Delaunay Triangulation:
Given a sets of points on 2D plane, find a
triangulation such that no points will strictly
inside circumcircle of any triangle. */
struct Edge {
    int id; // oidx[id]
    list<Edge>::iterator twin;
    Edge(int _id = 0):id(_id) {}
};
struct Delaunay { // 0-base
    int n, oidx[N];
    list<Edge> head[N]; // result udir. graph
    PT p[N];
    void init(int _n, PT _p[]) {
        n = _n, iota(oidx, oidx + n, 0);
        for (int i = 0; i < n; ++i) head[i].clear();
        sort(oidx, oidx + n, [&](int a, int b) {
            return _p[a] < _p[b];
        });
        for (int i = 0; i < n; ++i) p[i] = _p[oidx[i]];
    }
}

```

```

    divide(0, n - 1);
}
void addEdge(int u, int v) {
    head[u].push_front(Edge(v));
    head[v].push_front(Edge(u));
    head[u].begin()->twin = head[v].begin();
    head[v].begin()->twin = head[u].begin();
}
void divide(int l, int r) {
    if (l == r) return;
    if (l + 1 == r) return addEdge(l, l + 1);
    int mid = (l + r) >> 1, nw[2] = {l, r};
    divide(l, mid), divide(mid + 1, r);
    auto gao = [&](int t) {
        PT pt[2] = {p[nw[0]], p[nw[1]]};
        for (auto it : head[nw[t]]) {
            int v = ori(pt[1], pt[0], p[it.id]);
            if (v > 0 || (v == 0 &&
                (pt[t ^ 1] - p[it.id]) *
                (pt[t ^ 1] - p[it.id]) <
                (pt[1] - pt[0]) * (pt[1] - pt[0])))
                return nw[t] = it.id, true;
        }
        return false;
    };
    while (gao(0) || gao(1));
    addEdge(nw[0], nw[1]); // add tangent
    while (true) {
        PT pt[2] = {p[nw[0]], p[nw[1]]};
        int ch = -1, sd = 0;
        for (int t = 0; t < 2; ++t)
            for (auto it : head[nw[t]])
                if (ori(pt[0], pt[1], p[it.id]) > 0 &&
                    (ch == -1 || InCir(pt[0], pt[1],
                        p[ch], p[it.id])))
                    ch = it.id, sd = t;
        if (ch == -1) break; // upper common tangent
        for (auto it = head[nw[sd]].begin();
            it != head[nw[sd]].end(); )
            if (HasStrictInter(pt[sd], p[it->id],
                pt[sd ^ 1], p[ch]))
                head[it->id].erase(it->twin),
                head[nw[sd]].erase(it++);
            else ++it;
        nw[sd] = ch, addEdge(nw[0], nw[1]);
    }
}
} tool;

```

8.17 Voronoi Diagram [c6bcb2]

```

// (int) needs delaunay triangulation
// all coord is even, may call HalfPlaneInter after
vector<vector<Line>> vec;
void BuildVoronoiLine(int n, PT *arr) {
    tool.init(n, arr); // Delaunay
    vec.clear(), vec.resize(n);
    for (int i = 0; i < n; ++i)
        for (auto e : tool.head[i]) {
            int u = tool.oidx[i], v = tool.oidx[e.id];
            PT m((arr[v].x + arr[u].x) / 2,
                (arr[v].y + arr[u].y) / 2);
            PT d = ccw90(arr[v] - arr[u]);
            vec[u].push_back({m, m + d});
        }
}

```

8.18 3D Basic [4a7a37]

```

// (FP) add Point*scalar for proj
struct Point { double x = 0, y = 0, z = 0; };
Point operator-(Point p1, Point p2)
{return Point{p1.x-p2.x, p1.y-p2.y, p1.z-p2.z};}
Point operator+(Point p1, Point p2)
{return Point{p1.x+p2.x, p1.y+p2.y, p1.z+p2.z};}
Point operator/(Point p1, double v)
{return Point{p1.x / v, p1.y / v, p1.z / v};}
Point operator^(Point p1, Point p2) {
    return Point{p1.y * p2.z - p1.z * p2.y,
        p1.z * p2.x - p1.x * p2.z,
        p1.x * p2.y - p1.y * p2.x};
}

```

```

double operator*(Point p1, Point p2)
{ return p1.x * p2.x + p1.y * p2.y + p1.z * p2.z; }
double abs(Point a) { return sqrt(a * a); }
Point cross3(Point a, Point b, Point c)
{ return (b - a) ^ (c - a); }
double area2(Point a, Point b, Point c)
{ return abs(cross3(a, b, c)); }
double volume(Point a, Point b, Point c, Point d)
{ return cross3(a, b, c) * (d - a); }
Point Masscenter(Point a, Point b, Point c, Point d)
{ return (a + b + c + d) / 4; }
PT proj(Point a, Point b, Point c, Point u) {
    // proj. u to the plane of a, b, and c
    Point e1 = b - a;
    Point e2 = c - a;
    e1 = e1 / abs(e1);
    e2 = e2 - e1 * (e2 * e1);
    e2 = e2 / abs(e2);
    Point p = u - a;
    return PT{p * e1, p * e2};
}

```

8.19 3D Convex Hull [166108]

```

// (FP) needs 3D basic
// add Point==Point; returns {} unless hull is 3D
struct Face {
    int a, b, c;
    Face(int _a, int _b, int _c) : a(_a), b(_b), c(_c) {}
};
auto preprocess(auto pts) {
    auto G = pts.begin();
    vector<int> id;
    auto fail = tuple{-1, -1, -1, id};
    int a = int(find_if(iter(pts), [&](Point z) {
        return z != *G;
    }) - G);
    if (a == sz(pts)) return fail;
    int b = int(find_if(iter(pts), [&](Point z) {
        return cross3(*G, pts[a], z) != Point{0, 0, 0};
    }) - G);
    if (b == sz(pts)) return fail;
    int c = int(find_if(iter(pts), [&](Point z) {
        return sign(volume(*G, pts[a], pts[b], z)) != 0;
    }) - G);
    if (c == sz(pts)) return fail;
    for (int i = 0; i < sz(pts); i++)
        if (i != a && i != b && i != c)
            id.push_back(i);
    return tuple{a, b, c, id};
}
// return the faces with pts indexes
vector<Face> ConvexHull3D(vector<Point> pts) {
    int n = sz(pts);
    if (n <= 3) return {}; // be careful about edge case
    vector<Face> now;
    vector<vector<int>> z(n, vector<int>(n));
    auto [a, b, c, ord] = preprocess(pts);
    if (a == -1) return {};
    now.emplace_back(a, b, c); now.emplace_back(c, b, a);
    for (auto i : ord) {
        vector<Face> nxt;
        for (auto &f : now) {
            auto v = volume(pts[f.a], pts[f.b],
                pts[f.c], pts[i]);
            if (sign(v) <= 0) nxt.push_back(f);
            z[f.a][f.b] = z[f.b][f.c] =
                z[f.c][f.a] = sign(v);
        }
        auto F = [&](int x, int y) {
            if (z[x][y] > 0 && z[y][x] <= 0)
                nxt.emplace_back(x, y, i);
        };
        for (auto &f : now)
            F(f.a, f.b), F(f.b, f.c), F(f.c, f.a);
        now = nxt;
    }
    return now;
}
// n^2 delaunay: facets with negative z normal of
// convexhull of (x, y, x^2 + y^2), use a pseudo-point
// (0, 0, inf) to avoid degenerate case

```

```
// test @ SPOJ CH3D
// double area = 0, vol = 0; // surface area / volume
// for (auto [a, b, c]: faces)
//     area += abs(ver(p[a], p[b], p[c]))/2.0,
//     vol += volume(P3(0, 0, 0), p[a], p[b], p[c])/6.0;
```

9 Misc

9.1 All LCS [ba9cc9]

```
void all_lcs(string s, string t) { // 0-base
    vector<int> h(sz(t));
    iota(all(h), 0);
    for (int a = 0; a < sz(s); ++a) {
        int v = -1;
        for (int c = 0; c < sz(t); ++c)
            if (s[a] == t[c] || h[c] < v)
                swap(h[c], v);
        // LCS(s[0, a], t[b, c]) =
        // c - b + 1 - sum([h[i] >= b] | i <= c)
        // h[i] might become -1 !!
    }
}
```

9.2 Binary Search On Fraction [765c5a]

```
struct Q {
    ll p, q;
    Q go(Q b, ll d) { return {p + b.p*d, q + b.q*d}; }
};
bool pred(Q);
// returns smallest p/q in [lo, hi] such that
// pred(p/q) is true, and 0 <= p,q <= N
Q frac_bs(ll N) {
    Q lo{0, 1}, hi{1, 0};
    if (pred(lo)) return lo;
    assert(pred(hi));
    bool dir = 1, L = 1, H = 1;
    for (; L || H; dir = !dir) {
        ll len = 0, step = 1;
        for (int t = 0; t < 2 && (t ? step/=2 : step*=2);)
            if (Q mid = hi.go(lo, len + step);
                mid.p > N || mid.q > N || dir ^ pred(mid))
                t++;
            else len += step;
        swap(lo, hi = hi.go(lo, len));
        (dir ? L : H) = !!len;
    }
    return dir ? hi : lo;
}
```

9.3 Cyclic Ternary Search [9017cc]

```
/* bool pred(int a, int b);
f(0) ~ f(n - 1) is a cyclic-shift U-function
return idx s.t. pred(x, idx) is false forall x*/
int cyc_tsearch(int n, auto pred) {
    if (n == 1) return 0;
    int l = 0, r = n; bool rv = pred(1, 0);
    while (r - l > 1) {
        int m = (l + r) / 2;
        if (pred(0, m) ? rv : pred(m, (m + 1) % n)) r = m;
        else l = m;
    }
    return pred(1, r % n) ? l : r % n;
}
```

9.4 Matroid Intersection

$M = (E, \mathcal{I})$, where $\mathcal{I} \subseteq 2^E$ is nonempty, is a matroid if:

- If $S \in \mathcal{I}$ and $S' \subsetneq S$, then $S' \in \mathcal{I}$.
- For $S_1, S_2 \in \mathcal{I}$ s.t. $|S_1| < |S_2|$, there exists $e \in S_2 \setminus S_1$ s.t. $S_1 \cup \{e\} \in \mathcal{I}$.

Matroid intersection:

Start from $S = \emptyset$. In each iteration, let

- $Y_1 = \{x \notin S \mid S \cup \{x\} \in \mathcal{I}_1\}$
- $Y_2 = \{x \notin S \mid S \cup \{x\} \in \mathcal{I}_2\}$

If there exists $x \in Y_1 \cap Y_2$, insert x into S . Otherwise for each $x \in S, y \notin S$, create edges

- $x \rightarrow y$ if $S - \{x\} \cup \{y\} \in \mathcal{I}_1$.
- $y \rightarrow x$ if $S - \{x\} \cup \{y\} \in \mathcal{I}_2$.

Find a *shortest* path (with BFS) starting from a vertex in Y_1 and ending at a vertex in Y_2 which doesn't pass through any other vertices in Y_2 , and alternate the path. The size of S will be incremented by 1 in each iteration. For the weighted case, assign weight $w(x)$ to vertex x if $x \in S$ and $-w(x)$ if $x \notin S$. Find the path with the minimum number of edges among all minimum length paths and alternate it.

9.5 Min Plus Convolution [081e9d]

```
// a is convex a[i+1]-a[i] <= a[i+2]-a[i+1]
vector<int> min_plus_convolution(vector<int> &a, vector
<int> &b) {
    int n = sz(a), m = sz(b);
    vector<int> c(n + m - 1, INF);
    auto dc = [&](auto Y, int l, int r, int jl, int jr) {
        if (l > r) return;
        int mid = (l + r) / 2, from = -1, &best = c[mid];
        for (int j = jl; j <= jr; ++j)
            if (int i = mid - j; i >= 0 && i < n)
                if (best > a[i] + b[j])
                    best = a[i] + b[j], from = j;
        Y(Y, l, mid - 1, jl, from), Y(Y, mid + 1, r, from,
            jr);
    };
    return dc(dc, 0, n - 1 + m - 1, 0, m - 1), c;
}
```

9.6 Mo's Algorithm [ea5261]

```
struct MoAlgorithm {
    struct query {
        int l, r, id;
        bool operator < (const query &o) {
            if (l / C == o.l / C)
                return (l / C) & 1 ? r > o.r : r < o.r;
            return l / C < o.l / C;
        }
    };
    int cur_ans;
    vector<int> ans;
    void add(int x) {} // do something
    void sub(int x) {} // do something
    vector<query> Q;
    void add_query(int l, int r, int id) { // [L, R]
        Q.push_back({l, r, id});
        ans.push_back(0);
    }
    void run() {
        sort(Q.begin(), Q.end());
        int pl = 0, pr = 0;
        cur_ans = 0;
        for (query &i : Q) {
            while (pl > i.l) add(a[--pl]);
            while (pr < i.r) add(a[pr++]);
            while (pl < i.l) sub(a[pl++]);
            while (pr > i.r) sub(a[--pr]);
            ans[i.id] = cur_ans;
        }
    }
};
```

9.7 Mo's Algorithm On Tree [8331c2]

```
/* Mo's Algorithm On Tree
Preprocess:
1) LCA
2) dfs with in[u] = dft++, out[u] = dft++
3) ord[in[u]] = ord[out[u]] = u
4) bitset<MAXN> inset */
struct Query {
    int L, R, Lbid, lca;
    Query(int u, int v) {
        int c = LCA(u, v);
        if (c == u || c == v)
            q.lca = -1, q.L = out[c ^ u ^ v], q.R = out[c];
        else if (out[u] < in[v])
            q.lca = c, q.L = out[u], q.R = in[v];
        else
            q.lca = c, q.L = out[v], q.R = in[u];
        q.Lid = q.L / blk;
    }
};
```

```

bool operator<(const Query &q) const {
    if (LBid != q.LBid) return LBid < q.LBid;
    return R < q.R;
}
};
void flip(int x) {
    if (inset[x]) sub(arr[x]); // TODO
    else add(arr[x]); // TODO
    inset[x] = ~inset[x];
}
void solve(vector<Query> query) {
    sort(ALL(query));
    int L = 0, R = 0;
    for (auto q : query) {
        while (R < q.R) flip(ord[++R]);
        while (L > q.L) flip(ord[--L]);
        while (R > q.R) flip(ord[R--]);
        while (L < q.L) flip(ord[L++]);
        if (~q.lca) add(arr[q.lca]);
        // answer query
        if (~q.lca) sub(arr[q.lca]);
    }
}

```

9.8 PBDS [d65996]

```

#include <ext/pb_ds/tree_policy.hpp>
#include <ext/pb_ds/assoc_container.hpp>
using namespace __gnu_pbds;
tree<int, null_type, less<int>, rb_tree_tag,
    tree_order_statistics_node_update> oset;
// order_of_key, find_by_order
cc_hash_table<int, int> m1;
gp_hash_table<int, int> m2;
// Like map, but much faster

```

9.9 Simulated Annealing [de78c6]

```

double factor = 100000;
const int base = 1e9; // remember to run ~ 10 times
for (int it = 1; it <= 1000000; ++it) {
    // ans: answer, nw: current value, rnd(): mt19937
    if (exp(-(nw - ans) / factor) >= (double)rnd() %
        base) / base) ans = nw;
    factor *= 0.99995;
}

```

9.10 SOS dp [6aadb1]

```

//memory optimized, super easy to code.
rep(i, (1 << N)) F[i] = A[i];
rep(i, N) rep(mask, (1 << N)) {
    if (mask & (1<<i)) F[mask] += F[mask^(1<<i)];
}

```

9.11 SMAWK [e3ace9]

```

// For all 2x2 submatrix:
// If M[1][0] < M[1][1], M[0][0] < M[0][1]
// If M[1][0] == M[1][1], M[0][0] <= M[0][1]
// M[i][ans_i] is the best value in the i-th row
// select(int r, int u, int v) return true if f(r, v)
// is better than f(r, u)
vector<int> smawk(int N, int M, auto &&select) {
    auto dc = [&](auto self, const vector<int> &r, const
        vector<int> &c) {
        if (r.empty()) return vector<int>{};
        const int n = sz(r); vector<int> ans(n), nr, nc;
        for (int i : c) {
            while (!nc.empty() &&
                select(r[nc.size() - 1], nc.back(), i))
                nc.pop_back();
            if (int(nc.size()) < n) nc.push_back(i);
        }
        for (int i = 1; i < n; i += 2) nr.push_back(r[i]);
        const auto na = self(self, nr, nc);
        for (int i = 1; i < n; i += 2) ans[i] = na[i >> 1];
        for (int i = 0, j = 0; i < n; i += 2) {
            ans[i] = nc[j];
            const int end = i + 1 == n ? nc.back() : ans[i +
                1];
            while (nc[j] != end)

```

```

        if (select(r[i], ans[i], nc[++j])) ans[i] = nc[
            j];
        }
        return ans;
    };
    vector<int> R(N), C(M); iota(iter(R), 0), iota(iter(C)
        ), 0);
    return dc(dc, R, C);
}

```

9.12 Tree Hash [34aae5]

```

ull seed;
ull shift(ull x) {
    x ^= x << 13;
    x ^= x >> 7;
    x ^= x << 17;
    return x;
}
ull dfs(int u, int f) {
    ull sum = seed;
    for (int i : G[u]) if (i != f)
        sum += shift(dfs(i, u));
    return sum;
}

```

9.13 Python [ebfb5e]

```

from [decimal, fractions, math, random] import *
setcontext(Context(prec=10, Emax=MAX_EMAX, rounding=
    ROUND_FLOOR))
Decimal('1.1') / Decimal('0.2')
Fraction(3, 7)
Fraction(Decimal('1.14'))
Fraction('1.2').limit_denominator(4).numerator
Fraction(cos(pi / 3)).limit_denominator()
print(*[randint(1, C) for i in range(0, N)], sep=' ')

```